Phase 2b: repetition across broadcasts (shared stock, calibration, thrift per slot)

Generated by `analysis/parry/make_report.py` from `analysis/parry/results/`; rerun everything with `bash analysis/parry/run_all.sh`. The plan `analysis/parry/plan.md` was committed before any 2b test script was written or run (commit 34478c3 2026-10-08 00:16:36 +0000). **All of Phase 2b is exploratory relative to the pre-registered Phase 2**: it was added after every Phase 2 result had been seen, on the same corpus. Within 2b, seven tests were fixed in the plan as **2b-primary** and are Holm-corrected together (section 5); every other number is exploratory, and p-values outside section 5 are unadjusted. Brackets are 95% intervals; the kind of interval is stated each time.

Revision 1. This version adds the analyses of plan addendum 1 (dated and **post hoc**: written after the first report and the critic's review `review/critic_parry_v1.md`, whose own reruns the addendum adopts; its designs were committed before they were run, its implementation notes, including some verdict wording rules, after the first outputs; section 7 lists the rules and when each was fixed). The pre-registered analyses are rerun unchanged and reproduce: all 43 pre-registered results files are identical to commit 476a088 (run-time fields excluded; `results/rerun_check.csv`). The cross-corpus comparison now rests on 100,000-token identification sets (section 2.0); the 1,500-token matched design is kept as the secondary design. No verdict sentence in this report is typed by hand: each is chosen by a rule stated in `make_report.py` from the results files. Post hoc changes are listed in section 7.

Revision 2. Plan addendum 2 (dated and **post hoc**: written after the critic's second review `review/critic_parry_v2.md` and before its code was written) adds the rates of the TV-only strings at matched sample size (`p13_rate_ratio.py`), the slot composition of the TV-only stock corrected for the slot classifier's confusions (`p14_slot_adjust.py`), a same-slam / same-player sensitivity of the TV advantage (`p15_sameslam.py`), a check of the umpire mask and a composition variant in `p08_largeI.py` (new rows and files only), and wording changes (section 7, items 12-17). No pre-registered output and no addendum-1 number changes.

What is measured. As in Phase 2, 'formula' below means an exact word n-gram repeated in at least two utterances of a text (a repetition statistic), not Parry's metrically conditioned formula; the calibration in section 3 measures how far that statistic agrees with judgements of formulaicity by two LLM coders working from Parry's definition. A TV 'stream' (called 'team' in the plan) is one match with one unnamed broadcaster; team, match and broadcaster are confounded, and every commentator hint is [unverified].

Key results

- **Is there a TV-specific multiword stock? (post hoc; section 2.0).** Each TV stream (whole text) was covered by the formulas of a 100,000-token sample of the other 19 TV streams, of the Cornell live text and of the press answers (5 samples each; names and numerals normalised). Observed coverage, mean over the 20 targets: n >= 2: TV 59.7%, Cornell 44.6%, press 46.6%; n >= 3: TV 30.1%, Cornell 18.6%, press 18.7%. TV minus Cornell: 15.14 [14.29, 15.98] pp at n >= 2 (20 of 20 targets positive) and 11.43 [10.70, 12.16] pp at n >= 3 (20/20); TV minus press: 13.09 [11.85, 14.22] and 11.33 [10.36, 12.23] pp (20/20, 20/20; bootstrap over targets). TV sources of the same size cover TV targets more than written live text and press answers do, at n >= 2 and at n >= 3, for every one of the 20 targets. With the umpire's calls and point-score calls removed from the targets the n >= 3 differences are 11.08 [10.42, 11.79] and 10.85 [9.96, 11.68] pp; on raw tokens 10.63 [9.91, 11.38] and 8.06 [7.29, 8.78] pp. Differences of the excess over the unigram-shuffled null: n >= 2 10.27 / 5.42, n >= 3 12.60 / 10.27 pp (Cornell / press). Part of the advantage is sharing within a slam or around a player, which the baselines cannot have (post hoc, Table 2.H; all sources 30,000 tokens): without the other streams of the same slam and those sharing a player, the TV source's n >= 3 coverage falls from 16.6% to 15.7%, which removes 14.8 [9.7, 19.7]% of the gap to

Cornell and 13.4 [8.4, 18.2]% of the gap to press (ratio of means, bootstrap over targets); dropping as many streams at random removes -0.2 [-2.2, 1.7]% and -0.2 [-1.9, 1.7]%. The restricted TV source still covers TV targets more than Cornell (4.9 [4.5, 5.3] pp, 20/20 targets) and more than press (5.5 [4.9, 6.0] pp, 20/20 targets).

- **The TV-specific n >= 3 stock at equal sample sizes (post hoc; Tables 2.C-2.D).** In the 2019 final 10.3% of tokens lie in n >= 3 strings that the 100,000-token TV sample repeats and that neither the Cornell nor the press sample of the same size repeats ('TV-only'), and 20.7% in n >= 3 strings that a baseline sample also repeats (2023 final: 11.0% / 21.5%; mean over the 20 targets 9.6%, range 7.8-11.0%). Almost all of the TV-only tokens (8.9% of tokens on average, against 9.6%) are in strings attested in at least two other TV streams.
- **Rarer elsewhere, not absent (post hoc; Table 2.F2).** At matched sample size (100 samples of 100,000 tokens per source) a TV-only string's rate in TV is a median 8.4 times its rate in the more frequent baseline (occurrence-weighted quartiles 4.4-26.0; bootstrap interval over strings [7.8, 9.2]; the two halves of the samples give 8.1 and 8.2; against Cornell alone 5.7, against press alone 16.3; for the 300 commonest strings 15.4, quartiles 8.7-38.0), over the 84% of TV-only occurrences whose strings a baseline attests. 78% of those occurrences are in strings that the baseline uses less than once per 100,000 tokens (expected count below 1 in a sample of the design's size; Cornell 70%, press 86%), and 16% are in strings that neither baseline attests at all. The TV-only stock is therefore register-specific by rate (typically more than 8-fold), not by absolute absence. The ratio describes a selected stock: the strings were chosen for being repeated in a TV sample and not in the baseline samples, and their TV rate is measured on the text they were chosen from (strings shared with a baseline sample, chosen by the same TV criterion, have a median ratio of 0.9). The asymmetric reading, which tests a 100,000-token TV sample against the whole baselines (1.8 and 54 times larger), leaves 1.8% of tokens on average (2.8% in 2019) in strings that neither whole baseline repeats ('strict', Table 2.D); it is kept as a lower bound on the TV-only share, not as the measure of what the other registers use.
- **What the TV-only stock consists of (post hoc; Tables 2.E-2.G).** By the span-mode slot classifier (kappa 0.49 / 0.48 against the two coders; OTHER is its residual class), SCORE, STAT and OFFICIAL frames are 23% of the TV-only tokens, SHOT and JUDGE 46% and OTHER 29%. Corrected for the classifier's confusions on the coders' spans (Table 2.E2), SCORE, STAT and OFFICIAL are 38% [34, 41] (coder A) and 39% [35, 43] (coder B) and OTHER 5% / 5%; this correction carries P(coder slot | classifier slot) over from the coders' spans and returns about their own OTHER rate for every class (4-5%). The alternative correction, which carries over the classifier's error rates given the coder slot, gives 26% [13, 35] and 26% [15, 37], with OTHER 32% / 48% and wide intervals. SCORE, STAT and OFFICIAL frames are therefore a minority of the TV-only stock pooled over the 20 targets under every reading; their exact share, and the size of OTHER, are not determined by the calibration data. In the 2019 final they are 41% of the TV-only tokens as classified and 50% / 50% after the calibration correction (2023 final: 32%, 43% / 45%). 6% of the TV-only tokens lie inside umpire or score-call patterns (18% and 17% in the 2019 and 2023 finals), 2.3% inside the umpire's game announcements (`<name> leads by N games to M`, `game <name>`, `N games all`), which the classifier files under SCORE and which the commentary-only rows already exclude (32 of the 42 `leads (by) N games/sets to M` frames in the TV texts lie wholly inside the mask). Among the 40 most frequent TV-only strings, 20 are SCORE, STAT or OFFICIAL by modal slot (`and <num> minutes`, `<name> leads by`, `<num> miles an`, `<num> miles an hour`, `miles an hour`, `ball was called`), the others SHOT, JUDGE or OTHER (`he's trying to`, `that second serve`, `games to two`, `he's got to`, `by the way`, `look at this`; the classifier puts some score frames in OTHER, e.g. `games to two`, `six games to`).
- **Shared stock beyond lexicon (H1; a check that the pipeline detects collocation).** At 1,500 tokens per stream the formulas of one TV stream cover 8.7% of another's tokens against 1.1% for unigram-shuffled texts: excess 7.6 pp [7.0, 8.2] (n >= 2; 20/20 targets) and 1.02 pp [0.87, 1.18] (n >= 3). The shuffle destroys every collocation, and most of this excess is reached by sources that are not TV commentary: press answers reach 72% and Cornell live text 76% of it, and four press speakers share 10.9 pp among themselves (TV streams 7.6). H1 therefore

establishes repetition beyond the lexicon, not a commentary-specific stock; the TV-specific part is in the bullets above.

- **H2 at 1,500 tokens (pre-registered).** The pre-registered statistic (difference of excess over the shuffled null) is 1.84 pp [1.42, 2.33] against Cornell and 2.14 pp [1.66, 2.71] against press (Holm-rejected: yes, yes). On observed coverage, on which the claim rests, the differences are 0.53 [0.12, 1.01] pp (15/20 targets positive) and 3.10 [2.46, 3.84] pp (20/20), so at this size TV sources cover TV targets more than the written source (6% of the TV -> TV coverage of 8.70%) and more than the press source (36% of the TV -> TV coverage of 8.70%). The excess exceeds the observed difference for Cornell because the shuffled Cornell inventory is larger (54 types at 1,500 tokens against 16-32 for TV streams): Cornell's unigram distribution is more concentrated (Simpson index 0.0156 against 0.0096 per TV stream; <name> 5.6% and <num> 2.8% of tokens against 2.6% and 2.1%), so its chance bigrams cover shuffled TV text, and the size of the excess difference is a property of the null. At n >= 3 the 1,500-token inventories are near-empty (17-49 n >= 3 types per TV stream, TV -> TV coverage 1.02%) and their differences are not informative: the TV minus Cornell n >= 3 difference is -0.04 pp at 1,500 tokens, 0.15 at 3,000 and 11.43 at 100,000.
- **Genre core.** 8 n-grams are formulas of all 20 streams and 176 of at least 10 (22 of them with n >= 3; 0 with n >= 3 in all 20). Of the >= 10-stream core, 93% are formulas of a Cornell sample and 89% of a press sample of the same total size (120,682 tokens): most of the most widely shared strings are also formulas of the written and press baselines.
- **Stream-specific share (called 'idiolect' in the plan).** On full texts 17.0% of a stream's formula tokens (mean; range 11.8-20.0%) lie only in formulas that no other stream's inventory contains; at 1,500 tokens the share is 28.4% (range 24.0-31.9%). By slot (full texts) it is lowest for SCORE (10.6%) and highest for CROWD (23.6%; descriptive: the slot of a formula is computed from its own words). The share correlates with stream length (Spearman rho 0.44, p = 0.0510 on full texts; -0.66, p = 0.0015 at 1,500 tokens), and a stream is one match with one broadcaster, so the share mixes team and match topic and depends on sampling.
- **Clusters (exploratory).** Pairs from the same slam share more (excess 0.65 pp above other pairs; permutation p = 1.0e-04), which mixes broadcaster with venue and surface vocabulary. The only commentator-hint cluster with two members (the 2019 and 2023 Wimbledon finals, 'Tim') ranks 44 of 190 pairs (p = 0.2316): no sign that the hinted pair shares more than other pairs. Inventory overlap rises with shared players (Mantel r = 0.29, p = 0.0008).
- **Is the 2019 figure typical?** With the other 19 streams as identification set, 2019's coverage (raw tokens, n >= 2) ranks 13 of 20 (median 57.3%, range 50.2-64.4%); the Phase 2 value with the 18-stream pool is reproduced (55.03%).
- **Calibration (2b.2; the coders are two LLM agents).** The coders agree with each other at token-level kappa 0.77 [0.75, 0.80]. The automatic measures agree with them less than half as well: kappa 0.13-0.22 for the six measures of the brief against every coder reference set (0.11-0.22 with the exploratory measures); against the union of the coders precision exceeds its chance level (the coders' positive share, 0.28) by 0.07-0.16 and recall exceeds its chance level (the measure's own positive share) by 0.10-0.17. The Phase 2 measure (pool n >= 2) marks 54% of tokens: precision 0.35 (chance 0.28, ceiling 0.52), recall 0.68 (chance 0.54), kappa 0.16 [0.14, 0.18]; pool n >= 3: kappa 0.15, precision 0.39, recall 0.35 (chance 0.25); systems: kappa 0.22, precision 0.41, recall 0.55. Every automatic measure is therefore a weak proxy for the coders' judgement (kappa below 0.40 against every reference set). The higher precision on the 2019 utterances (0.43 vs 0.27 for coder A) is mostly a base rate (precision minus chance 0.09 vs 0.06): chance precision is 0.34 vs 0.21 (median utterance 26 vs 59 tokens; kappa 0.19 vs 0.12). At span level, 64% of coder A's HIGH spans lie wholly inside pool n >= 2 coverage (chance, by circular shifts: 35%) and 89% touch it (chance 79%).
- **Slot classifier.** In span mode (used for formula spans: Tables 2.11, 2.E-2.G, 4.7) it agrees with the coders' slot labels on 58% / 57% of their spans (kappa 0.49 / 0.48); its OTHER class is a residual that takes many of the coders' JUDGE and SCORE spans (Table 3.4), which biases compositions (Table 2.E2). In context mode (used for the situational slot of a reference in H4) it agrees with the coders on 46% / 49% of

the commentary references that lie inside a coder span (kappa 0.35 [0.22, 0.49] / 0.39 [0.25, 0.52], bootstrap over references; n = 65 / 69; the coder slot of a reference is that of the longest coder span covering it, a proxy for a judgement of the reference's situation). Slot labels this noisy (kappa below 0.6) attenuate any association between slot and form (H4b) toward zero and make per-slot attributions unreliable.

- **Naming habits (H4a).** Over 2833 commentary-only references to the players, the bare surname is 73%, the first name 12%, the full name 7%, hypocoristics 5% and descriptive epithets 3% (at most 4% if every unresolved epithet referred to a player). Streams (team, match and broadcaster together) differ in naming habits: 303 distinct stream x slot x category cells against 388.4 expected (p = 1.0e-04; surname 33-98% and first name 0-41% of references by stream). Verdict after the post hoc checks: **rejected (robust)** (p 1.0e-05-5.0e-05 over 8 seeds and variants, among them surname, first and full name only: p = 5.0e-05; leave-one-stream-out p <= 0.0002).
- **Slot-bound form choice (H4b).** Pre-registered: 472 distinct stream x player x slot forms against 483.3 expected, p = 0.0221, Holm p = 0.0442 (rejected). Verdict after the post hoc checks: **not robust / inconclusive**. With 100,000 permutations at three seeds p = 0.0233-0.0236 (Holm 0.0465-0.0473); without hypocoristic and epithet forms p = 0.0646; surname and first name only p = 0.1232; with the span-mode slot p = 0.0055 (span mode lets the reference's own words enter the classifier, e.g. an epithet containing `number one` counts as STAT, so this variant can favour rejection); without the 3 streams with hypocoristics p = 0.0443; one reference per player per utterance p = 0.0097; leaving out one stream at a time p = 0.0104-0.0902 (17 of 20 below 0.05). Per stream, 2 of 20 situational and 3 of 20 syntactic tests reach p < 0.05 (1 expected each). The transcripts have no speaker labels, so a play-by-play voice and an analyst voice that differ both in naming and in typical slot would produce the same deficit; H4b cannot separate that from slot-conditioned choice. No stream shows the Parryan pattern of one form per slot with different forms in different slots (situational slots: not one form per slot: 18 streams; one form throughout (economy without slot conditioning): 2 streams); the economy that exists is a stream's habitual naming, chiefly the surname.
- **Functional equivalents (E2, exploratory).** Of 16 tested equivalence classes (17 declared; the others have one attested form), stream choice among the alternatives is more consistent than chance (unadjusted p < 0.05, fewer forms per stream) for speed_unit (p = 1.0e-04), along_the_line (p = 0.0018), match_point (p = 0.0207), serve_wide (p = 0.0381); not for tied_at_forty, break_chance, net_error, serve_centre, serve_body, small_degree, praise_stroke, well_played, obligation, opinion_hedge, major_titles, audience. Of the 32 E2 p-values 10 are below 0.05 (1.6 expected); after a Bonferroni bound over the 32, only speed_unit remains.
- **Stream-level reuse per slot (E1, exploratory; called 'economy' in the plan).** With the inventory identified on all 20 streams pooled, every slot shows fewer distinct formula types per stream than chance; this is guaranteed by construction, because a string repeated inside one stream only enters the pooled inventory. With the inventory identified on the other 19 streams (post hoc), 7 of 7 slots still do (SCORE, OFFICIAL, SHOT, JUDGE, STAT, CROWD, OTHER): streams reuse their own selection of strings that other streams also use (a stream being one match and one broadcaster); Table 4.7b. Within-document burstiness (a match's player frames, a broadcaster's speed unit) produces this deficit in any collection of documents, so it is stream-level reuse, not evidence of economy in Parry's sense; no yardstick collection (e.g. Cornell per match) was run.
- **Extension (H5).** No association was detected between the syllable length of a reference and the time available after its clip: weighted within-stream rho = 0.028 (stream bootstrap [-0.012, 0.068]; permutation p = 0.1402; 2811 references in 20 streams; minimum detectable rho about 0.053); with the stored time field, valid in all 20 streams since the corpus correction, rho = 0.025, p = 0.1891 (post hoc).
- **2b-primary family (Holm, 7 tests; section 5).** Rejected as pre-registered: H1a, H1b, H2a, H2b, H4a, H4b; not rejected: H5. Read with the post hoc checks: H1a/H1b: most of the detected excess is also reached by the baseline sources; H2a/H2b on observed coverage: 0.53 [0.12, 1.01] and 3.10 [2.46, 3.84] pp (both

above 0), 15.1 and 13.1 pp with 100,000-token sources; H4a: rejected (robust); H4b: not robust / inconclusive; H5: not rejected.

1. Data and definitions

Teams (plan section 0). Utterance = one rally clip (TV, `text_corrected`), one live-text update (Cornell), one player answer (press).

team	medium	utterances	tokens
2019 WIM M F Federer-Djokovic	tv	318	9,791
2023 WIM M F Alcaraz-Djokovic	tv	245	7,216
2019 AO W F Osaka-Kvitova	tv	187	6,527
2020 AO M F Djokovic-Thiem	tv	262	8,371
2020 RG W F Swiatek-Kenin	tv	49	1,597
2021 AO W F Osaka-Brady	tv	112	3,276
2021 RG M F Tsitsipas-Djokovic	tv	228	7,837
2021 USO W F Raducanu-Fernandez	tv	115	3,498
2022 AO M F Nadal-Medvedev	tv	302	11,505
2022 RG M QF Djokovic-Nadal	tv	222	8,046
2022 USO M F Ruud-Alcaraz	tv	124	6,231
2023 RG W F Swiatek-Muchova	tv	173	3,800
2023 USO M F Djokovic-Medvedev	tv	141	5,992
2024 AO M F Sinner-Medvedev	tv	252	9,256
2024 RG W F Swiatek-Paolini	tv	86	2,368
2024 WIM W F Paolini-Krejcikova	tv	133	3,825
2024 USO M F Fritz-Sinner	tv	128	5,756
2025 AO M SF Sinner-Shelton	tv	176	7,113
2025 WIM W F Anisimova-Swiatek	tv	71	2,592
2025 USO M F Sinner-Alcaraz	tv	169	6,085
Cornell live text (written)	text	3962	178,770
press answers, pooled	press	81974	5,435,611
press: Federer	press	4192	430,223
press: Djokovic	press	4579	377,847
press: Murray	press	4265	356,166
press: Nadal	press	3425	286,973

Press answers are dated 2007-01-15 to 2015-10-18 (358 interviewees); the Cornell records carry no dates (outlet: Sports Mole live text (per Cornell dataset README; written, not speech) [source site not independently verified]); the TV matches are from 2019 to 2025.

Tokens. Phase 2 tokeniser and STOP list (`analysis/formulas/common.py`, unchanged). **2b normalisation** (primary): digit strings and `love fifteen thirty forty deuce` -> `<num>`; first names and surnames of the players of the 20 TV matches (Phase 2 lexicon), plus the players of a Cornell update and the interviewee of a press answer -> `<name>`. Raw Phase 2 tokens are a sensitivity. Hypocoristics (`rafa`, `nole`, `rog`, `carlitos`) are not in the name lexicon: 155 such tokens in the TV streams (1.28 per 1,000), 0.25 per 1,000 in press and 0.10 in Cornell. `<name>` is 2.60% of TV tokens, 5.59% of Cornell and 0.10% of press tokens (press answers rarely name the players of the TV matches), so frames around player names (`from <name>`, `for <name>`) cannot be shared from press for a reason of genre.

Formula of a team = n-gram (2 <= n <= 12) in >= 2 distinct utterances of the team's text, not made only of STOP words; variants n >= 3 and 'content' (not made only of STOP words and numerals). **Coverage** of a text by an inventory = share of its tokens inside an occurrence of an inventory n-gram. **Large identification sets** (post hoc, addendum 1): each TV stream (whole text) is the target; the source is a 100,000-token sample (whole utterances) of the other 19 TV streams, of Cornell or of press, 5 samples each; the target and each source sample are also unigram-shuffled once per sample (null). **Matched size** (pre-registered, now secondary): each team subsampled to S = 1,500 tokens (all 20 TV

streams) or 3,000 (17 TV streams), R = 500 / 200 replicates. **Null (a)**: tokens permuted over the positions of a text (lexicon, topic and utterance lengths kept, multiword units destroyed), identification and coverage repeated. **Excess** = observed minus shuffled coverage. **Vertex bootstrap**: the 20 TV streams resampled with replacement, the statistic averaged over ordered pairs of distinct original streams (each pair's replicate mean held fixed).

Situational slot classifier (plan section 5; code `lib2b.SlotClassifier`): (1) OFFICIAL if the span touches an umpire/Hawk-Eye pattern (`mr <name>`, `... is challenging`, `challenges remaining`, `ball was called`, `new balls please`, `thank you (players|please)`, `quiet please`, `time violation`, `hawk eye`, ...); (2) SCORE if it touches a score pattern (`<score> <score|all>`, `deuce`, `advantage <name>`, `game <name>`, `<name> leads by N games to M`, `N games all`, `break/set/match point`, `tie break`); (3) otherwise the slot whose keyword lexicon has most hits in the span (ties: OFFICIAL > SCORE > CROWD > STAT > SHOT > JUDGE); (4) the same on a window of 4 tokens either side; (5) OTHER. Two modes: **span mode** (`classify`, steps 1-5 on the span itself; used for formula spans) and **context mode** (`classify_context`, steps 1-3 on the 4-token window either side excluding the span; used for the situational slot of a reference, so that the name form cannot determine its own slot). Both modes are scored against the coders' slot labels in section 3. The lexicons are frozen in the plan and were not tuned on any 2b output.

Available time. 2b found that `t_to_next_first_hit_s` was invalid in 6 of the 20 streams (videos at about 29.97 frames per second whose clip times had been computed as frames/25); the corpus has since been corrected (corpus/README.md section 3.3: frame rate detected per stream, frame-derived times recomputed, hit times unchanged). The pre-registered analyses use the hit clock, which the correction did not touch: **A_after** = first hit of the next clip minus last hit of this clip (the clip's text runs into the following dead time). The stored field, valid in all 20 streams after the correction, enters a post hoc sensitivity of H5 (last row of Table 4.6).

2. Shared stock: what TV commentary repeats that other registers do not (2b.1)

2.0 Large identification sets (post hoc, addendum 1; the primary cross-corpus design of this revision)

Each of the 20 TV streams (whole text) is a target; a source is a 100,000-token sample of the other 19 TV streams pooled, of the Cornell live text or of the press answers (5 samples per source and target); its inventory is identified on the sample. Commentary-only = target tokens inside `common.official_mask_v2` patterns (umpire calls, Hawk-Eye announcements, point-score calls) removed from numerator and denominator. Intervals: percentile bootstrap over the 20 targets (B = 10,000 for differences). The targets' TV sources share 18 of their 19 streams, so this bootstrap treats targets as exchangeable and understates between-broadcast uncertainty; the number of targets with a positive difference is given for that reason.

Table 2.A. Coverage of TV targets by 100,000-token sources (% of the target's tokens, mean over the 20 targets [bootstrap 95% CI]; min over targets). A target's TV source is drawn from the other 19 streams (109,177-119,085 tokens), so each 100,000-token draw holds 84-92% of its pool and the 5 TV samples of a target nearly coincide; a Cornell draw holds 56% of Cornell and a press draw 1.8% of the press corpus. The intervals are over targets, so this overlap does not narrow them.

tokens	source	inventory types	of which n >= 3	observed n >= 2	observed n >= 3	shuffled n >= 2	shuffled n >= 3	commentary-only n >= 3	min n >= 3
norm	other 19 TV streams	18670	10029	59.7 [58.3, 61.2]	30.1 [28.9, 31.2]	30.5	1.28	29.6 [28.5, 30.7]	24.2
norm	Cornell live text	31559	23312	44.6 [42.8,	18.6 [17.5,	25.7	2.44	18.5 [17.4, 19.7]	14.2

tokens	source	inventory types	of which n >= 3	observed n >= 2	observed n >= 3	shuffled n >= 2	shuffled n >= 3	commentary-only n >= 3	min n >= 3
	(written)			46.4]	19.8]				
norm	press answers, pooled	20479	12365	46.6 [45.9, 47.4]	18.7 [18.0, 19.5]	22.9	0.22	18.8 [18.1, 19.5]	15.9
raw	other 19 TV streams	17798	8909	55.6 [54.0, 57.2]	25.9 [24.9, 26.9]	23.5	0.21	25.9 [24.8, 26.9]	20.5
raw	Cornell live text (written)	27979	18788	39.1 [37.3, 40.9]	15.3 [14.4, 16.2]	18.3	0.54	15.5 [14.6, 16.4]	11.0
raw	press answers, pooled	20306	12203	42.8 [41.9, 43.6]	17.9 [17.1, 18.6]	18.7	0.17	18.1 [17.4, 18.8]	14.5

Table 2.B. TV sources minus baseline sources of the same size (pp; mean over targets [bootstrap 95% CI]; targets with a positive difference; min over targets).

tokens	baseline	measure	TV minus baseline	targets > 0	min	reading (95% CI vs 0)
norm	Cornell live text (written)	observed n >= 2	15.14 [14.29, 15.98]	20/20	11.25	TV more
norm	Cornell live text (written)	observed n >= 3	11.43 [10.70, 12.16]	20/20	8.74	TV more
norm	Cornell live text (written)	commentary-only n >= 2	15.17 [14.31, 16.01]	20/20	11.25	TV more
norm	Cornell live text (written)	commentary-only n >= 3	11.08 [10.42, 11.79]	20/20	8.80	TV more
norm	Cornell live text (written)	shuffled n >= 2	4.86 [4.32, 5.41]	20/20	2.92	TV more
norm	Cornell live text (written)	shuffled n >= 3	-1.17 [-1.36, -1.00]	0/20	-2.10	TV less
norm	Cornell live text (written)	excess n >= 2	10.27 [9.78, 10.74]	20/20	7.97	TV more
norm	Cornell live text (written)	excess n >= 3	12.60 [11.97, 13.27]	20/20	10.78	TV more
norm	press answers, pooled	observed n >= 2	13.09 [11.85, 14.22]	20/20	5.75	TV more
norm	press answers, pooled	observed n >= 3	11.33 [10.36, 12.23]	20/20	6.55	TV more
norm	press answers, pooled	commentary-only n >= 2	12.98 [11.75, 14.13]	20/20	5.75	TV more
norm	press answers, pooled	commentary-only n >= 3	10.85 [9.96, 11.68]	20/20	6.37	TV more
norm	press answers, pooled	shuffled n >= 2	7.67 [6.85, 8.48]	20/20	3.23	TV more
norm	press answers, pooled	shuffled n >= 3	1.06 [0.92, 1.21]	20/20	0.52	TV more
norm	press answers, pooled	excess n >= 2	5.42 [4.74, 6.10]	20/20	2.52	TV more
norm	press answers, pooled	excess n >= 3	10.27 [9.41, 11.09]	20/20	5.71	TV more
raw	Cornell live text (written)	observed n >= 2	16.52 [15.56, 17.44]	20/20	13.09	TV more
raw	Cornell live text (written)	observed n >= 3	10.63 [9.91, 11.38]	20/20	8.06	TV more
raw	Cornell live text (written)	commentary-only n >= 2	16.15 [15.21, 17.08]	20/20	12.53	TV more
raw	Cornell live text (written)	commentary-only n >= 3	10.42 [9.72, 11.17]	20/20	8.09	TV more
raw	Cornell live text (written)	shuffled n >= 2	5.24 [4.59, 5.89]	20/20	2.49	TV more

tokens	baseline	measure	TV minus baseline	targets > 0	min	reading (95% CI vs 0)
raw	Cornell live text (written)	shuffled n >= 3	-0.33 [-0.39, -0.28]	0/20	-0.70	TV less
raw	Cornell live text (written)	excess n >= 2	11.28 [10.83, 11.73]	20/20	9.11	TV more
raw	Cornell live text (written)	excess n >= 3	10.96 [10.26, 11.70]	20/20	8.46	TV more
raw	press answers, pooled	observed n >= 2	12.82 [11.58, 13.99]	20/20	6.22	TV more
raw	press answers, pooled	observed n >= 3	8.06 [7.29, 8.78]	20/20	3.59	TV more
raw	press answers, pooled	commentary-only n >= 2	12.09 [10.90, 13.20]	20/20	5.83	TV more
raw	press answers, pooled	commentary-only n >= 3	7.76 [7.01, 8.45]	20/20	3.46	TV more
raw	press answers, pooled	shuffled n >= 2	4.81 [4.01, 5.61]	20/20	0.90	TV more
raw	press answers, pooled	shuffled n >= 3	0.04 [0.01, 0.08]	14/20	-0.09	TV more
raw	press answers, pooled	excess n >= 2	8.01 [7.31, 8.72]	20/20	5.32	TV more
raw	press answers, pooled	excess n >= 3	8.02 [7.27, 8.72]	20/20	3.68	TV more

Table 2.C. Per target (2b normalisation; observed coverage %, mean over the 5 samples; TV-only and shared n >= 3 shares as defined below).

target	tokens	n>=2 TV	n>=2 Cornell	n>=2 press	n>=3 TV	n>=3 Cornell	n>=3 press	n>=3 shared %	n>=3 TV-only %	TV-only strict %	TV-only, commentary-only %
2019 WIM M F Federer-Djokovic	9,791	60.7	46.7	47.3	31.0	19.5	18.2	20.7	10.3	2.8	8.8
2023 WIM M F Alcaraz-Djokovic	7,216	62.6	46.5	46.7	32.5	19.5	17.7	21.5	11.0	2.7	9.5
2019 AO W F Osaka-Kvitova	6,527	57.4	44.0	44.5	28.6	18.9	18.6	20.5	8.1	1.2	8.1
2020 AO M F Djokovic-Thiem	8,371	60.2	45.3	47.0	31.8	19.5	20.2	22.0	9.8	1.8	9.6
2020 RG W F Swiatek-Kenin	1,597	65.2	50.1	48.0	33.0	20.0	20.0	23.4	9.6	1.2	9.4
2021 AO W F Osaka-Brady	3,276	61.1	49.8	45.6	30.4	21.7	17.3	20.9	9.5	1.5	9.5
2021 RG M F Tsitsipas-Djokovic	7,837	61.2	48.9	46.3	31.2	21.7	16.8	21.3	9.8	2.0	9.6
2021 USO W F Raducanu-Fernandez	3,498	55.0	40.2	45.4	25.8	15.1	16.7	17.6	8.2	1.6	8.0

target	tokens	n>=2 TV	n>=2 Cornell	n>=2 press	n>=3 TV	n>=3 Cornell	n>=3 press	n>=3 shared %	n>=3 TV-only %	TV-only strict %	TV-only, commentary-only %
2022 AO M F Nadal-Medvedev	11,505	59.8	42.1	47.8	31.9	17.7	21.0	21.4	10.5	1.6	10.4
2022 RG M QF Djokovic-Nadal	8,046	61.8	45.6	46.7	31.3	20.6	18.0	22.1	9.2	1.8	9.1
2022 USO M F Ruud-Alcaraz	6,231	57.9	39.6	49.0	28.5	15.3	20.8	19.7	8.7	1.7	8.3
2023 RG W F Swiatek-Muchova	3,800	63.8	49.7	48.3	33.4	22.6	20.4	24.2	9.2	2.2	8.9
2023 USO M F Djokovic-Medvedev	5,992	60.5	43.4	47.7	30.6	17.7	19.1	20.0	10.6	1.3	10.2
2024 AO M F Sinner-Medvedev	9,256	58.4	40.9	45.9	30.3	16.4	18.9	19.7	10.5	1.3	10.3
2024 RG W F Swiatek-Paolini	2,368	66.1	52.2	49.4	33.0	24.1	21.0	23.1	9.9	2.1	10.0
2024 WIM W F Paolini-Krejcikova	3,825	58.2	46.1	44.0	29.6	18.4	18.1	19.1	10.5	2.3	9.9
2024 USO M F Fritz-Sinner	5,756	53.8	36.8	43.3	24.4	14.2	15.9	16.6	7.8	1.1	7.6
2025 AO M SF Sinner-Shelton	7,113	61.1	43.9	48.9	31.8	17.9	21.5	21.7	10.1	1.7	9.8
2025 WIM W F Anisimova-Swiatek	2,592	52.8	38.1	47.0	24.2	14.4	17.7	16.0	8.2	1.3	8.1
2025 USO M F Sinner-Alcaraz	6,085	57.2	42.1	44.1	28.2	17.4	17.0	18.9	9.3	1.9	8.8

TV-only n >= 3 stock (addendum 1). In each sample, an occurrence in the target of an n >= 3 string of the TV inventory is *shared* if the string is also in the Cornell or the press inventory of that sample, otherwise *TV-only*; a covered token is shared if any covering occurrence is shared. *Strict*: every covering TV-only string is also not a formula (>= 2 utterances) of the whole Cornell text or of the whole press corpus. *Cross-broadcast*: every covering TV-only string occurs in at least two of the other 19 TV streams. A TV-only token's slot is the span-mode slot of the longest covering TV-only occurrence.

Table 2.D. Shares of target tokens (%; mean over the 5 samples; mean and range over the 20 targets, and the two finals).

class	mean over 20 targets	range	2019 final (in-sample)	2023 final (held-out)
covered by an n >= 3 TV string	30.1	24.2-33.4	31.0	32.5
shared with a baseline	20.5	16.0-24.2	20.7	21.5

class	mean over 20 targets	range	2019 final (in-sample)	2023 final (held-out)
TV-only	9.6	7.8-11.0	10.3	11.0
TV-only, cross-broadcast	8.9	7.4-9.9	9.4	9.9
TV-only, strict (asymmetric: against the whole baselines)	1.8	1.1-2.8	2.8	2.7
not in the Cornell inventory	15.7	12.9-18.1	16.0	16.6
not in the press inventory	15.5	11.8-18.6	16.8	18.6
TV-only, commentary-only tokens	9.2	7.6-10.4	8.8	9.5
TV-only strict, commentary-only tokens	1.5	0.9-2.1	1.6	1.6

Table 2.E. Composition by situational slot, as classified (% of each class's tokens, span-mode slot of the longest covering occurrence). These are the classifier's outputs (kappa 0.49 / 0.48 against the two coders); OTHER is its residual class (no pattern and no keyword in the span or its window), which takes many spans the coders label JUDGE or SCORE (Table 3.4), so the shares are biased; Table 2.E2 corrects them. Last two rows: % inside umpire or score-call patterns (`official_mask_v2`) and inside the umpire's game announcements (`game <name>`, `<name> leads (by) N games to M`, `N games all`), which the frozen classifier files under SCORE; the last column labels those announcements OFFICIAL. Pooled over the 20 targets unless stated.

slot	TV-only	TV-only, cross-broadcast	TV-only, strict	shared	2019 final: TV-only	2023 final: TV-only	TV-only, umpire game calls as OFFICIAL
SCORE	9.3	9.4	15.9	8.8	14.3	14.3	7.0
OFFICIAL	4.6	4.0	11.3	1.0	13.4	10.5	6.9
SHOT	27.8	28.0	27.9	26.7	20.6	28.0	27.8
JUDGE	18.3	18.5	11.0	25.5	13.2	13.7	18.3
STAT	8.9	8.9	5.9	8.0	13.2	7.2	8.9
CROWD	1.8	1.7	2.6	1.1	2.3	2.5	1.8
OTHER	29.3	29.4	25.5	28.8	22.8	23.9	29.3
inside umpire/score-call patterns	5.8	5.4	17.6	2.1	18.1	17.2	5.8
inside the umpire's game announcements	2.3	2.3	8.7	0.1	7.0	7.6	2.3

Table 2.E2. Composition of the TV-only tokens corrected for the classifier's confusions (post hoc, addendum 2; `p14_slot_adjust.py`; %, pooled over the 20 targets [percentile bootstrap 95% CI over the calibration utterances and the targets]). The confusions are those of span mode on the coders' spans (Table 3.4; 1,249 / 1,010 spans of coders A / B, each span weighted equally). *Calibration* carries P(coder slot | classifier slot) over to the TV-only strings (the critic's correction); it returns roughly the coders' own slot mix wherever the classifier is unsure, e.g. about their OTHER rate for every class. *Inversion* carries over P(classifier slot | coder slot), the classifier's error rates, and solves for the composition (non-negative least squares, shares summing to 1); it does not borrow the coders' slot mix but is poorly conditioned. Both assume that confusions measured on coder formula spans hold for TV-only n-grams.

slot	classifier (Table 2.E)	calibration, coder A	calibration, coder B	inversion, coder A	inversion, coder B
SCORE	9.3 [7.8, 10.9]	19.2 [16.8, 21.6]	17.0 [14.7, 19.3]	10.1 [2.3, 15.7]	8.4 [2.9, 12.9]
OFFICIAL	4.6 [2.7, 6.7]	4.8 [3.0, 6.8]	4.9 [3.0, 7.0]	3.6 [0.0, 6.7]	1.4 [0.0, 6.2]
SHOT	27.8 [25.8, 30.3]	20.4 [17.8, 23.2]	21.8 [18.7, 25.2]	14.8 [6.0, 21.7]	0.0 [0.0, 15.4]
JUDGE	18.3 [16.6, 19.8]	33.7 [30.8, 36.5]	30.7 [27.4, 34.1]	23.7 [0.0, 36.1]	24.7 [5.7, 43.1]
STAT	8.9 [7.5, 10.3]	13.7 [11.4, 16.0]	17.2 [14.3, 20.3]	12.4 [5.8, 17.9]	16.2 [7.7, 25.4]

slot	classifier (Table 2.E)	calibration, coder A	calibration, coder B	inversion, coder A	inversion, coder B
CROWD	1.8 [1.5, 2.0]	3.5 [2.2, 5.1]	3.7 [2.0, 5.6]	3.3 [1.9, 6.0]	1.4 [0.0, 12.2]
OTHER	29.3 [27.4, 31.2]	4.7 [3.4, 6.2]	4.7 [3.1, 6.5]	32.1 [16.2, 62.1]	47.9 [15.6, 61.6]
SCORE+STAT+OFFICIAL	22.8 [19.2, 26.7]	37.7 [34.1, 41.4]	39.1 [35.2, 43.1]	26.1 [12.9, 35.4]	26.0 [15.3, 37.5]
SHOT+JUDGE	46.1 [43.1, 49.1]	54.1 [50.4, 57.9]	52.5 [48.1, 56.7]	38.6 [13.9, 52.1]	24.7 [7.9, 53.6]

SCORE + STAT + OFFICIAL under other weightings and classes (calibration, coder A / coder B): spans weighted by their length 37.2% / 39.2% (inversion 24.0% / 25.5%); cross-broadcast: classifier 22.4%, corrected 37.4% / 38.8%; strict: classifier 33.1%, corrected 44.7% / 46.3%; shared: classifier 17.9%, corrected 33.7% / 35.1%; 2019 final: classifier 41.0%, corrected 49.6% / 50.5%; 2023 final: classifier 31.9%, corrected 43.4% / 44.8%.

Table 2.F. The 40 commonest TV-only strings (pooled over targets: strings that are TV-only in a majority of the 5 samples of a target; occurrences in those targets; modal slot; TV streams attesting the string; rates per 100,000 tokens in the 20 TV streams, the whole Cornell text and the whole press corpus; strict = not repeated in either whole baseline).

string	targets	occurrences	modal slot	TV streams	per 100k TV	per 100k Cornell	per 100k press	strict
and <num> minutes	6	19	STAT	7	16.6	1.1	0.81	no
he's trying to	9	19	OTHER	9	15.7	0.0	0.59	no
<name> leads by	5	18	SCORE	5	14.9	0.0	0.00	yes
<num> miles an	6	18	STAT	7	18.2	0.0	0.64	no
<num> miles an hour	6	18	STAT	7	18.2	0.0	0.64	no
miles an hour	6	18	STAT	7	18.2	0.0	0.79	no
that second serve	11	18	SHOT	11	14.9	0.0	0.20	no
ball was called	4	17	OFFICIAL	4	14.1	0.0	0.09	no
hours and <num>	6	17	STAT	6	14.1	0.0	0.39	no
hours and <num> minutes	6	17	STAT	6	14.1	0.0	0.37	no
games to two	9	16	OTHER	9	13.3	0.0	0.00	yes
he's got to	5	16	JUDGE	9	23.2	1.1	1.09	no
games to one	7	15	SCORE	7	12.4	0.0	0.02	yes
by the way	7	14	OTHER	8	12.4	0.0	0.68	no
look at this	6	14	OTHER	6	11.6	0.0	0.33	no
on the left	8	14	OTHER	8	11.6	0.6	0.29	no
<name> <name> and	8	13	OTHER	8	10.8	0.6	0.53	no
challenging the call	2	13	OFFICIAL	2	10.8	0.0	0.02	yes
do you think	8	13	JUDGE	8	10.8	0.0	1.42	no
ladies and gentlemen	6	13	OFFICIAL	6	10.8	0.0	0.00	yes
look at that	9	13	OTHER	9	10.8	0.0	0.68	no
on that forehand	9	13	SHOT	9	10.8	0.0	0.09	no
on the outside	4	13	SHOT	4	10.8	0.6	0.94	no
that first serve	10	13	SHOT	10	10.8	0.0	0.20	no
the back end	6	13	SHOT	6	10.8	1.1	0.18	no

string	targets	occurrences	modal slot	TV streams	per 100k TV	per 100k Cornell	per 100k press	strict
the new balls	8	13	OFFICIAL	8	10.8	0.0	0.20	no
two games to	7	13	SCORE	7	10.8	0.0	0.11	no
back in <num>	7	12	OTHER	8	10.8	0.0	1.03	no
five games to	9	12	SCORE	9	9.9	0.0	0.02	yes
have a look	6	12	OTHER	6	9.9	0.0	0.85	no
i just wonder	6	12	OTHER	6	9.9	0.0	0.00	yes
six games to	7	12	OTHER	7	9.9	0.0	0.04	no
to the forehand	7	12	SHOT	7	9.9	0.0	0.57	no
with the new	7	12	OFFICIAL	7	9.9	0.0	0.63	no
<name> with the	7	11	OTHER	7	9.1	0.6	0.06	no
<num> <num> first	9	11	SCORE	9	9.1	0.0	0.24	no
<num> second set	7	11	SCORE	7	9.1	0.0	0.15	no
four games to	9	11	SCORE	9	9.1	0.0	0.04	no
from two sets	5	11	SCORE	5	9.1	0.0	0.66	no
has two challenges	5	11	OFFICIAL	5	9.1	0.0	0.00	yes

Names: 3 of these 40 strings contain <name>. Full names are a TV habit: the bigram <name> <name> occurs 239 times per 100,000 tokens in the TV streams, 65 in Cornell and 5.8 in the press answers, so strings such as <name> <name> and are TV-only partly through that habit and the name normalisation; the raw-token rows of Table 2.B show that the TV advantage does not depend on the normalisation.

Table 2.F2. Rates of the TV-only strings at matched sample size (post hoc, addendum 2; p13_rate_ratio.py). Rate = occurrences per 100,000 tokens, mean over 100 samples of 100,000 tokens per source (TV: the 20 streams pooled); ratio = TV rate / baseline rate over the strings the baseline attests ('max' = the more frequent baseline); median and quartiles weighted as stated; [bootstrap 95% CI over strings, descriptive]; halves = the median from samples 1-50 and 51-100. 'Whole' rows use the whole texts instead of samples.

strings	weighting	rates	baseline	attested strings (share of weight)	median ratio	quartiles	halves	% ratio >= 10	median baseline rate per 100k	% baseline rate < 1 per 100k	un i ba
TV-only, all (2,606 strings)	occurrences	matched	max	2152 (84%)	8.4 [7.8, 9.2]	4.4-26.0	8.1 / 8.2	45	0.53	78	16
TV-only, all (2,606 strings)	occurrences	matched	Cornell	774 (30%)	5.7 [5.4, 6.1]	4.1-8.5	5.7 / 5.8	17	0.59	70	16
TV-only, all (2,606 strings)	occurrences	matched	press	2028 (80%)	16.3 [14.4, 17.8]	6.2-42.8	14.7 / 15.0	62	0.25	86	16
TV-only, all (2,606 strings)	types	matched	max	2152 (83%)	6.0 [5.6, 6.6]	3.3-20.3	5.8 / 5.9	37	0.55	73	17
TV-only, all (2,606 strings)	occurrences	whole	max	2185 (85%)	8.9	4.4-27.0	-	46	0.56	78	15
TV-only, 300 commonest	occurrences	matched	max	252 (84%)	15.4 [11.9, 19.1]	8.7-38.0	14.3 / 14.3	66	0.54	82	16
TV-only, 300 commonest	types	whole	max	260 (87%)	12.4	8.1-39.4	-	61	0.56	83	13

strings	weighting	rates	baseline	attested strings (share of weight)	median ratio	quartiles	halves	% ratio >= 10	median baseline rate per 100k	% baseline rate < 1 per 100k	un i ba
shared with a baseline sample (1,940 strings)	occurrences	matched	max	1940 (100%)	0.9 [0.8, 1.0]	0.5-1.7	0.9 / 0.9	0	8.89	0	0

Formula status in equal-size samples (occurrence-weighted): a TV-only string is repeated (>= 2 utterances) in 97% of the TV samples, 3.5% of the Cornell samples and 7.8% of the press samples; a shared string in 99%, 65% and 63%. The ratios describe a selected stock (strings chosen for being repeated in a TV sample and not in the baseline samples, with their TV rate measured on the text they were chosen from). The 300 commonest strings, the set of the critic's check, give larger ratios than the whole stock; the critic's check corresponds to the row '300 commonest, types, whole' (quantiles here are lower weighted quantiles, so quartiles can differ slightly from interpolated ones).

Table 2.G. The commonest TV-only strings of the two finals (25 each; occurrences in the final; rate per 100,000 tokens in the other 19 TV streams).

final	string	occurrences	modal slot	other TV streams	per 100k TV	per 100k Cornell	per 100k press	strict
2019 WIM M F Federer-Djokovic	`<name> leads by`	7	SCORE	4	9.9	0.0	0.00	yes
2019 WIM M F Federer-Djokovic	`<name> is challenging`	5	OFFICIAL	1	1.8	0.0	0.00	yes
2019 WIM M F Federer-Djokovic	`and <num> minutes`	5	STAT	6	13.5	1.1	0.81	no
2019 WIM M F Federer-Djokovic	`hours and <num>`	5	STAT	5	10.8	0.0	0.39	no
2019 WIM M F Federer-Djokovic	`hours and <num> minutes`	5	STAT	5	10.8	0.0	0.37	no
2019 WIM M F Federer-Djokovic	`the semi final`	5	STAT	2	1.8	0.6	0.02	yes
2019 WIM M F Federer-Djokovic	`<name> has two challenges`	4	OFFICIAL	2	4.5	0.0	0.00	yes
2019 WIM M F Federer-Djokovic	`<name> has two challenges remaining`	4	OFFICIAL	2	4.5	0.0	0.00	yes
2019 WIM M F Federer-Djokovic	`<num> miles an`	4	STAT	6	16.2	0.0	0.64	no
2019 WIM M F Federer-Djokovic	`<num> miles an hour`	4	STAT	6	16.2	0.0	0.64	no

final	string	occurrences	modal slot	other TV streams	per 100k TV	per 100k Cornell	per 100k press	strict
2019 WIM M F Federer-Djokovic	ball was called	4	OFFICIAL	3	11.7	0.0	0.09	no
2019 WIM M F Federer-Djokovic	games to one	4	SCORE	6	9.9	0.0	0.02	yes
2019 WIM M F Federer-Djokovic	has two challenges	4	OFFICIAL	4	6.3	0.0	0.00	yes
2019 WIM M F Federer-Djokovic	has two challenges remaining	4	OFFICIAL	3	5.4	0.0	0.00	yes
2019 WIM M F Federer-Djokovic	in the <name>	4	OTHER	2	1.8	0.6	0.00	yes
2019 WIM M F Federer-Djokovic	miles an hour	4	STAT	6	16.2	0.0	0.79	no
2019 WIM M F Federer-Djokovic	on their feet	4	OTHER	2	1.8	2.2	0.09	no
2019 WIM M F Federer-Djokovic	please thank you	4	OFFICIAL	3	2.7	0.0	0.00	yes
2019 WIM M F Federer-Djokovic	thank you please	4	OFFICIAL	2	2.7	0.0	0.00	yes
2019 WIM M F Federer-Djokovic	two challenges remaining	4	OFFICIAL	3	5.4	0.0	0.00	yes
2019 WIM M F Federer-Djokovic	<name> <name> the	3	SCORE	2	1.8	0.0	0.06	no
2019 WIM M F Federer-Djokovic	<name> able to	3	SHOT	3	4.5	0.0	0.00	yes
2019 WIM M F Federer-Djokovic	<name> trying to	3	OTHER	4	4.5	0.0	0.02	yes
2019 WIM M F Federer-Djokovic	by two games	3	SCORE	3	4.5	0.0	0.00	yes
2019 WIM M F Federer-Djokovic	by two games to	3	SCORE	3	4.5	0.0	0.00	yes
2023 WIM M F Alcaraz-Djokovic	challenging the call	10	OFFICIAL	1	2.6	0.0	0.02	yes
2023 WIM M F	ball was called	7	OFFICIAL	3	8.8	0.0	0.09	no

final	string	occurrences	modal slot	other TV streams	per 100k TV	per 100k Cornell	per 100k press	strict
Alcaraz-Djokovic								
2023 WIM M F Alcaraz-Djokovic	ball was called out	5	OFFICIAL	3	4.4	0.0	0.07	no
2023 WIM M F Alcaraz-Djokovic	have a look	5	SHOT	5	6.2	0.0	0.85	no
2023 WIM M F Alcaraz-Djokovic	have a look at	5	SHOT	5	5.3	0.0	0.40	no
2023 WIM M F Alcaraz-Djokovic	was called out	5	OFFICIAL	3	5.3	1.1	0.13	no
2023 WIM M F Alcaraz-Djokovic	<name> has two challenges	4	OFFICIAL	2	4.4	0.0	0.00	yes
2023 WIM M F Alcaraz-Djokovic	<name> has two challenges remaining	4	OFFICIAL	2	4.4	0.0	0.00	yes
2023 WIM M F Alcaraz-Djokovic	<name> leads five	4	SCORE	2	1.8	0.0	0.00	yes
2023 WIM M F Alcaraz-Djokovic	<name> leads five games	4	SCORE	2	1.8	0.0	0.00	yes
2023 WIM M F Alcaraz-Djokovic	<name> leads five games to	4	SCORE	2	1.8	0.0	0.00	yes
2023 WIM M F Alcaraz-Djokovic	five games to	4	SCORE	8	7.1	0.0	0.02	yes
2023 WIM M F Alcaraz-Djokovic	has two challenges	4	OFFICIAL	4	6.2	0.0	0.00	yes
2023 WIM M F Alcaraz-Djokovic	has two challenges remaining	4	OFFICIAL	3	5.3	0.0	0.00	yes
2023 WIM M F Alcaraz-Djokovic	leads five games	4	SCORE	3	2.6	0.0	0.00	yes
2023 WIM M F Alcaraz-Djokovic	leads five games to	4	SCORE	3	2.6	0.0	0.00	yes
2023 WIM M F Alcaraz-Djokovic	look at this	4	OTHER	5	8.8	0.0	0.33	no
2023 WIM M F Alcaraz-Djokovic	two challenges remaining	4	OFFICIAL	3	5.3	0.0	0.00	yes

final	string	occurrences	modal slot	other TV streams	per 100k TV	per 100k Cornell	per 100k press	strict
2023 WIM M F Alcaraz-Djokovic	`<num> miles an`	3	STAT	6	16.7	0.0	0.64	no
2023 WIM M F Alcaraz-Djokovic	`<num> miles an hour`	3	STAT	6	16.7	0.0	0.64	no
2023 WIM M F Alcaraz-Djokovic	`first and second`	3	SHOT	3	6.2	0.6	2.21	no
2023 WIM M F Alcaraz-Djokovic	`games to one`	3	SCORE	6	10.6	0.0	0.02	yes
2023 WIM M F Alcaraz-Djokovic	`look at that`	3	OTHER	8	8.8	0.0	0.68	no
2023 WIM M F Alcaraz-Djokovic	`miles an hour`	3	STAT	6	16.7	0.0	0.79	no
2023 WIM M F Alcaraz-Djokovic	`new balls please`	3	OFFICIAL	3	2.6	0.0	0.00	yes

Table 2.H. Same slam and shared players (post hoc, addendum 2; `p15_sameslam.py`). Every source subsampled to 30,000 tokens (5 samples per target; the smallest restricted pool has 31,098 tokens); observed coverage of the 20 TV targets (2b normalisation; % of tokens, mean over targets [bootstrap 95% CI over targets]); streams in the pool: min-max over targets.

source	n >= 2	n >= 3	streams in pool
(a) other 19 TV streams	43.6 [42.2, 44.8]	16.6 [15.9, 17.3]	19-19
(b) other streams, different slam	42.9 [41.5, 44.2]	16.0 [15.4, 16.6]	14-16
(c) other streams, no shared player	43.0 [41.7, 44.3]	16.3 [15.6, 17.0]	12-19
(d) different slam and no shared player	42.5 [41.2, 43.8]	15.7 [15.1, 16.4]	8-15
(e) control: as many streams removed at random	43.4 [42.1, 44.6]	16.6 [15.8, 17.3]	8-15
(f) Cornell live text	32.4 [30.9, 33.9]	10.8 [10.1, 11.6]	-
(g) press answers	31.6 [30.9, 32.2]	10.3 [9.8, 10.7]	-

Share of the TV-minus-baseline gap removed by each restriction (%; (a - x) / (a - baseline), ratio of means over targets [bootstrap 95% CI]):

restriction	n >= 3, gap to Cornell	n >= 3, gap to press	n >= 2, gap to Cornell	n >= 2, gap to press
different slam (b)	10.0 [5.8, 14.1]	9.1 [5.1, 13.1]	5.9 [3.6, 8.2]	5.5 [3.3, 7.7]
no shared player (c)	5.3 [3.6, 7.2]	4.8 [3.3, 6.5]	4.8 [2.7, 7.1]	4.5 [2.7, 6.3]
both (d)	14.8 [9.7, 19.7]	13.4 [8.4, 18.2]	9.4 [7.2, 11.3]	8.7 [6.4, 11.0]
random control (e)	-0.2 [-2.2, 1.7]	-0.2 [-1.9, 1.7]	1.6 [-0.2, 3.4]	1.5 [-0.2, 3.2]
both, beyond the random control (e - d)	14.9 [10.1, 19.9]	13.6 [9.2, 17.9]	7.8 [5.5, 9.9]	7.2 [4.9, 9.4]

The two finals with sources of 54,000 tokens (n >= 3 coverage %, mean of 5 samples): 2019 WIM M F Federer-Djokovic: a 23.3, b 22.7, c 22.0, d 21.7, e 22.8, f 14.8, g 13.8; 2023 WIM M F Alcaraz-Djokovic: a 24.4, b 23.8, c 23.9, d 23.0, e 23.6, f 14.8, g 13.3; the restriction (d) removes 17% of the gap to Cornell and 15% of the gap to press (the two finals pooled; random control 7% / 7%).

2.1 Matched small size (pre-registered design; secondary)

Table 2.1. Inventory sizes (formula types; full texts, and mean over replicates at matched size; 2b normalisation). The press corpus (5.4 million tokens) is not indexed whole.

team	full: n>=2	full: n>=3	full: n>=2 per 1,000 tokens	1,500: n>=2	1,500: n>=3	1,500: n>=2 shuffled	3,000: n>=2
2019 WIM M F Federer-Djokovic	1288	532	131.5	95	25	32	262
2023 WIM M F Alcaraz-Djokovic	1031	428	142.9	110	35	26	308
2019 AO W F Osaka-Kvitova	723	231	110.8	93	21	19	254
2020 AO M F Djokovic-Thiem	1117	455	133.4	95	26	19	264
2020 RG W F Swiatek-Kenin	108	28	67.6	101	26	23	-
2021 AO W F Osaka-Brady	360	122	109.9	114	31	31	321
2021 RG M F Tsitsipas-Djokovic	1065	404	135.9	108	30	26	294
2021 USO W F Raducanu-Fernandez	284	74	81.2	86	19	17	232
2022 AO M F Nadal-Medvedev	1429	560	124.2	86	22	18	240
2022 RG M QF Djokovic-Nadal	1073	415	133.4	99	24	25	275
2022 USO M F Ruud-Alcaraz	743	275	119.2	94	25	18	268
2023 RG W F Swiatek-Muchova	471	176	123.9	128	36	25	343
2023 USO M F Djokovic-Medvedev	636	194	106.1	88	18	23	249
2024 AO M F Sinner-Medvedev	1146	446	123.8	86	20	17	237
2024 RG W F Swiatek-Paolini	251	78	106.0	130	38	29	-
2024 WIM W F Paolini-Krejcikova	537	248	140.4	128	49	19	381
2024 USO M F Fritz-Sinner	490	150	85.1	74	17	18	205
2025 AO M SF Sinner-Shelton	995	404	139.9	104	30	21	292
2025 WIM W F Anisimova-Swiatek	188	52	72.5	82	20	18	-
2025 USO M F Sinner-Alcaraz	639	227	105.0	81	18	18	233
Cornell live text (written)	59323	46228	331.8	198	73	53	504
press answers, pooled	-	-	-	91	26	19	245
press: Federer	-	-	-	92	28	18	254
press: Djokovic	-	-	-	104	33	17	291
press: Murray	-	-	-	121	41	24	325
press: Nadal	-	-	-	117	37	27	326

Table 2.2. Coverage of TV targets by each kind of source, and of the baselines by TV sources (% of the target's tokens; mean over pairs; 95% CI from the bootstrap over TV streams). Excess = observed - shuffled.

S	inventory	source -> target	observed	shuffled	excess
1500	n >= 2	TV -> TV	8.70 [8.04, 9.46]	1.14	7.56 [7.00, 8.21]
1500	n >= 2	Cornell -> TV	8.17 [7.59, 8.77]	2.45	5.72 [5.36, 6.10]
1500	n >= 2	press -> TV	5.60 [5.30, 5.92]	0.18	5.42 [5.12, 5.73]
1500	n >= 2	TV -> Cornell	9.14 [8.61, 9.67]	2.50	6.64 [6.35, 6.98]
1500	n >= 2	TV -> press	7.03 [6.63, 7.43]	0.22	6.82 [6.45, 7.24]
1500	n >= 3 (near-empty inventories: not informative)	TV -> TV	1.02 [0.87, 1.18]	0.00	1.02 [0.88, 1.18]
1500	n >= 3 (near-empty inventories: not informative)	Cornell -> TV	1.06 [0.95, 1.18]	0.01	1.05 [0.94, 1.16]
1500	n >= 3 (near-empty inventories: not informative)	press -> TV	1.04 [0.93, 1.15]	0.00	1.04 [0.93, 1.15]
1500	n >= 3 (near-empty inventories: not informative)	TV -> Cornell	0.74 [0.63, 0.85]	0.00	0.74 [0.63, 0.85]
1500	n >= 3 (near-empty inventories: not informative)	TV -> press	1.08 [0.97, 1.20]	0.00	1.08 [0.97, 1.20]
1500	content n >= 2	TV -> TV	6.96 [6.32, 7.71]	0.73	6.23 [5.66, 6.92]
1500	content n >= 2	Cornell -> TV	6.17 [5.69, 6.64]	1.76	4.41 [4.09, 4.75]
1500	content n >= 2	press -> TV	4.94 [4.63, 5.28]	0.13	4.81 [4.52, 5.15]
1500	content n >= 2	TV -> Cornell	6.19 [5.70, 6.70]	1.83	4.36 [4.06, 4.66]
1500	content n >= 2	TV -> press	6.44 [6.06, 6.87]	0.11	6.33 [5.93, 6.74]
3000	n >= 2	TV -> TV	14.59 [13.77, 15.46]	2.64	11.94 [11.28, 12.68]
3000	n >= 2	Cornell -> TV	12.50 [11.68, 13.31]	4.24	8.26 [7.74, 8.78]
3000	n >= 2	press -> TV	9.52 [9.09, 9.99]	0.60	8.92 [8.54, 9.32]
3000	n >= 2	TV -> Cornell	15.84 [15.02, 16.64]	5.26	10.57 [10.13, 11.04]
3000	n >= 2	TV -> press	12.06 [11.51, 12.60]	0.62	11.44 [10.92, 11.98]
3000	n >= 3	TV -> TV	2.38 [2.14, 2.62]	0.00	2.37 [2.15, 2.62]
3000	n >= 3	Cornell -> TV	2.22 [1.98, 2.47]	0.04	2.18 [1.93, 2.43]
3000	n >= 3	press -> TV	1.90 [1.72, 2.08]	0.00	1.90 [1.72, 2.07]
3000	n >= 3	TV -> Cornell	1.98 [1.72, 2.24]	0.01	1.96 [1.71, 2.23]
3000	n >= 3	TV -> press	2.32 [2.12, 2.49]	0.00	2.32 [2.13, 2.50]
3000	content n >= 2	TV -> TV	12.36 [11.61, 13.19]	1.71	10.66 [10.01, 11.37]
3000	content n >= 2	Cornell -> TV	10.30 [9.57, 11.04]	3.02	7.27 [6.81, 7.75]
3000	content n >= 2	press -> TV	8.08 [7.63, 8.57]	0.41	7.68 [7.26, 8.13]
3000	content n >= 2	TV -> Cornell	12.15 [11.38, 12.86]	3.87	8.28 [7.81, 8.68]
3000	content n >= 2	TV -> press	11.16 [10.64, 11.69]	0.35	10.81 [10.28, 11.33]

Table 2.3. TV sources minus baseline sources at matched size (pp; vertex bootstrap 95% CI, B = 10,000; targets with a positive difference). The pre-registered H2 statistic is the excess column (n >= 2, S = 1,500: H2a, H2b); the observed column is the basis of the H2 claim (addendum 1, M1). The excess intervals here come from new bootstrap draws (`p09_h2_observed.py`) and can differ from section 5's in the second decimal.

S	inventory	baseline	observed difference (targets > 0)	shuffled difference	excess difference (targets > 0)
1500	n >= 2	Cornell	0.53 [0.12, 1.01] (15/20)	-1.31 [-1.36, -1.27]	1.84 [1.43, 2.32] (20/20)
1500	n >= 2	press	3.10 [2.46, 3.84] (20/20)	0.96 [0.73, 1.23]	2.14 [1.67, 2.71] (20/20)
1500	n >= 3 (near-empty: not informative)	Cornell	-0.04 [-0.22, 0.15] (10/20)	-0.01 [-0.01, -0.01]	-0.03 [-0.21, 0.16] (10/20)
1500	n >= 3 (near-empty: not informative)	press	-0.02 [-0.12, 0.10] (8/20)	0.00 [0.00, 0.00]	-0.02 [-0.12, 0.10] (8/20)
1500	content n >= 2	Cornell	0.79 [0.36, 1.29] (17/20)	-1.03 [-1.08, -0.99]	1.82 [1.41, 2.28] (20/20)
1500	content n >= 2	press	2.02 [1.43, 2.70] (20/20)	0.60 [0.42, 0.81]	1.42 [0.95, 1.98] (20/20)
3000	n >= 2	Cornell	2.09 [1.62, 2.59] (17/17)	-1.60 [-1.68, -1.50]	3.68 [3.22, 4.19] (17/17)
3000	n >= 2	press	5.06 [4.28, 5.96] (17/17)	2.04 [1.60, 2.58]	3.02 [2.50, 3.62] (17/17)
3000	n >= 3	Cornell	0.15 [-0.18, 0.49] (10/17)	-0.03 [-0.04, -0.03]	0.19 [-0.15, 0.53] (10/17)
3000	n >= 3	press	0.48 [0.32, 0.67] (17/17)	0.00 [0.00, 0.01]	0.47 [0.31, 0.67] (17/17)
3000	content n >= 2	Cornell	2.07 [1.56, 2.63] (17/17)	-1.32 [-1.38, -1.25]	3.38 [2.91, 3.91] (17/17)
3000	content n >= 2	press	4.28 [3.49, 5.17] (17/17)	1.30 [0.94, 1.71]	2.98 [2.45, 3.56] (17/17)

Table 2.3b. Why the shuffled Cornell inventory is large (2b normalisation; 50 shuffled 1,500-token samples per team for the inventory columns).

corpus	<name> %	<num> %	Simpson sum p^2	top-10 types %	observed inventory at 1,500	shuffled inventory at 1,500	shuffled types with a placeholder %
TV, 20 streams pooled	2.60	2.09	0.0089	24.2	-	-	-
TV, mean over streams	2.62	2.10	0.0096	24.7	99.1	22.3 (16-32)	53
Cornell	5.59	2.75	0.0156	30.7	193.1	54.4	59
press, pooled	0.10	0.57	0.0107	25.2	91.9	19.0	4

Table 2.4. Jaccard overlap of inventories (mean over pairs, %; observed, shuffled, excess).

S	inventory	pair type	observed	shuffled	excess
1500	n >= 2	TV - TV	8.77 [8.36, 9.20]	4.46	4.31
1500	n >= 2	Cornell - TV	5.00 [4.68, 5.36]	5.95	-0.94
1500	n >= 2	press - TV	6.16 [5.78, 6.56]	0.34	5.82
1500	n >= 3	TV - TV	2.30 [1.92, 2.72]	-	-
1500	n >= 3	Cornell - TV	0.98 [0.82, 1.14]	-	-
1500	n >= 3	press - TV	2.35 [2.04, 2.66]	-	-
3000	n >= 2	TV - TV	10.67 [10.12, 11.18]	7.51	3.16
3000	n >= 2	Cornell - TV	6.20 [5.82, 6.60]	7.58	-1.37
3000	n >= 2	press - TV	7.03 [6.64, 7.41]	1.04	5.99
3000	n >= 3	TV - TV	3.61 [3.18, 4.07]	-	-

S	inventory	pair type	observed	shuffled	excess
3000	n >= 3	Cornell - TV	1.75 [1.48, 2.01]	0.18	1.56
3000	n >= 3	press - TV	2.78 [2.47, 3.07]	-	-

'-' = undefined (both shuffled inventories empty). Coverage, not Jaccard, carries the tests; Table 2.3b gives the inventory sizes behind the Cornell rows.

Table 2.5. Raw-token sensitivity (no <num>/<name> normalisation; S = 1,500).

inventory	comparison	observed	excess
n >= 2	TV -> TV	6.01 [5.47, 6.64]	5.85 [5.33, 6.47]
n >= 2	Cornell -> TV	4.87 [4.60, 5.16]	4.51 [4.25, 4.77]
n >= 2	press -> TV	5.27 [4.97, 5.60]	5.13 [4.83, 5.45]
n >= 2	TV minus Cornell live text (written)	1.14 [0.68, 1.70]	1.34 [0.88, 1.89]
n >= 2	TV minus press answers, pooled	0.74 [0.31, 1.26]	0.72 [0.31, 1.21]
n >= 3	TV -> TV	0.93 [0.78, 1.10]	0.93 [0.78, 1.09]
n >= 3	Cornell -> TV	0.83 [0.74, 0.92]	0.83 [0.74, 0.92]
n >= 3	press -> TV	1.03 [0.92, 1.14]	1.03 [0.92, 1.14]
n >= 3	TV minus Cornell live text (written)	0.10 [-0.08, 0.29]	0.10 [-0.08, 0.29]
n >= 3	TV minus press answers, pooled	-0.10 [-0.20, 0.03]	-0.10 [-0.20, 0.02]

Table 2.6. Single-speaker press baselines (S = 1,500, n >= 2; coverage of one interviewee's answers by another's inventory, observed / excess, %).

source \ target	press: Federer	press: Djokovic	press: Murray	press: Nadal
press: Federer	-	10.3 / 9.9	13.8 / 13.2	9.2 / 8.8
press: Djokovic	10.6 / 10.2	-	13.9 / 13.4	10.2 / 9.7
press: Murray	12.2 / 11.5	12.2 / 11.6	-	11.2 / 10.7
press: Nadal	9.8 / 9.3	10.5 / 10.0	12.7 / 12.1	-

Mean excess between press speakers: 10.86 pp, against 7.56 pp between TV streams (press speakers share more among themselves). The yardstick is not matched: a speaker's 1,500 tokens come from answers given over many years (topic-diverse, so their repeats are generic) in edited transcripts, a TV stream's from one match in raw ASR.

Table 2.7. Each TV stream as target and as source (S = 1,500, n >= 2, 2b normalisation; % ; mean over the other 19 TV streams).

stream	tokens	as target: observed	as target: excess	as source: excess	Cornell -> it: excess	press -> it: excess
2019 WIM M F Federer-Djokovic	9,791	9.94	8.15	6.78	6.43	5.20
2023 WIM M F Alcaraz-Djokovic	7,216	9.60	8.15	7.58	6.37	4.85
2019 AO W F Osaka-Kvitova	6,527	7.81	6.84	7.79	5.51	4.54
2020 AO M F Djokovic-Thiem	8,371	8.98	8.07	8.07	5.35	6.45
2020 RG W F Swiatek-Kenin	1,597	9.59	8.51	8.16	5.76	6.21
2021 AO W F Osaka-Brady	3,276	9.68	8.12	7.55	6.04	4.91
2021 RG M F Tsitsipas-Djokovic	7,837	8.91	7.50	7.64	6.51	4.55
2021 USO W F Raducanu-Fernandez	3,498	7.97	7.07	7.10	4.83	5.19
2022 AO M F Nadal-Medvedev	11,505	8.85	8.03	7.44	5.41	6.76
2022 RG M QF Djokovic-Nadal	8,046	8.56	7.26	7.58	6.80	5.11
2022 USO M F Ruud-Alcaraz	6,231	7.90	6.96	7.14	4.89	5.48

stream	tokens	as target: observed	as target: excess	as source: excess	Cornell -> it: excess	press -> it: excess
2023 RG W F Swiatek-Muchova	3,800	10.63	9.46	9.30	7.52	6.36
2023 USO M F Djokovic-Medvedev	5,992	8.12	6.99	7.46	5.80	5.02
2024 AO M F Sinner-Medvedev	9,256	8.34	7.40	7.91	4.90	6.13
2024 RG W F Swiatek-Paolini	2,368	10.15	8.71	8.22	7.03	5.92
2024 WIM W F Paolini-Krejcikova	3,825	8.75	7.76	8.01	5.63	5.30
2024 USO M F Fritz-Sinner	5,756	7.09	6.04	6.52	4.13	4.65
2025 AO M SF Sinner-Shelton	7,113	8.43	7.38	7.61	5.25	6.15
2025 WIM W F Anisimova-Swiatek	2,592	7.37	6.42	6.33	5.14	5.10
2025 USO M F Sinner-Alcaraz	6,085	7.35	6.38	6.98	5.02	4.59

2.2 Genre core

Table 2.8. Core size: formula types in the inventories of at least k of the 20 TV streams (full texts; and mean over 200 subsamples of 1,500 tokens).

k	full: n>=2	full: n>=3	full: content	full raw: n>=2	1,500: n>=2	1,500: n>=3
2	2118	683	1958	1966	269.7	49.5
3	1181	325	1076	1063	138.3	18.3
4	797	196	724	703	88.7	9.1
5	561	128	505	490	64.1	5.4
6	430	85	386	371	49.5	3.6
7	335	58	300	286	38.8	2.6
8	264	41	233	221	30.8	2.0
9	215	30	189	176	24.7	1.5
10	176	22	154	143	20.0	1.1
11	137	17	119	109	16.3	0.7
12	115	11	101	90	13.4	0.3
13	84	5	75	65	11.4	0.1
14	74	5	65	55	9.2	0.0
15	57	5	53	46	7.0	0.0
16	50	5	46	41	5.1	0.0
17	42	4	39	34	3.5	0.0
18	29	2	27	22	2.2	0.0
19	19	0	17	16	1.4	0.0
20	8	0	6	6	0.8	0.0

Table 2.9. The most widely shared formulas (full texts; K = number of streams whose inventory contains it; modal span-mode slot over its occurrences).

formula	n	K	utterances (all streams)	modal slot	share
`<num> <num>`	2	20	591	SCORE	73%
`a little`	2	20	303	JUDGE	49%
`for <name>`	2	20	240	OTHER	39%
`going to`	2	20	218	OTHER	43%
`the first`	2	20	215	OTHER	33%
`the ball`	2	20	191	OTHER	46%
`the court`	2	20	154	OTHER	61%
`so far`	2	20	72	OTHER	31%
`i think`	2	19	245	JUDGE	100%

formula	n	K	utterances (all streams)	modal slot	share
from <name>	2	19	238	SHOT	39%
to get	2	19	188	OTHER	47%
a lot	2	19	151	OTHER	43%
able to	2	19	117	OTHER	46%
trying to	2	19	114	OTHER	50%
first serve	2	19	110	SHOT	100%
second set	2	19	94	OTHER	50%
a good	2	19	93	JUDGE	100%
couple of	2	19	93	OTHER	34%
first set	2	19	84	OTHER	48%
<name> <name>	2	18	245	OTHER	41%
<name> is	2	18	162	OTHER	38%
as well	2	18	148	OTHER	50%
the line	2	18	135	SHOT	99%
that <name>	2	18	119	OTHER	50%
lot of	2	18	102	OTHER	31%
a lot of	3	18	98	SHOT	31%
and <name>	2	18	93	SHOT	34%
a few	2	18	74	OTHER	34%
of the court	3	18	72	OTHER	67%
little bit	2	17	164	JUDGE	100%
a little bit	3	17	160	JUDGE	100%
the second	2	17	121	OTHER	45%
i mean	2	17	118	OTHER	43%
thank you	2	17	117	OFFICIAL	100%
this match	2	17	98	OTHER	56%
got to	2	17	87	JUDGE	100%
to go	2	17	80	OTHER	35%
a couple	2	17	79	OTHER	27%
<name> was	2	17	76	OTHER	41%
to take	2	17	76	SHOT	34%

Table 2.10. Is the TV core shared with the baselines? Share of the core's types (full texts) that are formulas of a Cornell or press random sample of 120,682 tokens (the size of the 20 TV streams together; mean [2.5-97.5%] over 20 samples).

k	core types	in Cornell inventory	in press inventory	n>=3 core types	n>=3 in Cornell	n>=3 in press
2	2118	60.9 [60.2, 62.4]	62.5 [61.4, 63.3]	683	48.9	51.8
5	561	83.6 [82.5, 84.5]	83.5 [82.5, 85.3]	128	77.7	79.8
10	176	93.0 [92.0, 94.3]	88.7 [86.3, 90.6]	22	89.8	90.0
15	57	97.5 [96.5, 98.2]	93.9 [90.3, 96.5]	5	100.0	100.0
20	8	100.0 [100.0, 100.0]	95.0 [87.5, 100.0]	0	0.0	0.0

Table 2.11. Stream-specific ('idiolect') and core shares by situational slot (descriptive: the slot is the span-mode slot of the formula's own occurrence, so SCORE and OFFICIAL rows are partly defined by the formulas' words; share of formula tokens; full texts first, then 1,500-token subsamples, mean of 200, with min-max over streams).

slot	full: stream-specific %	full: core-10 %	formula tokens per stream (1,500)	1,500: stream-specific % (range)	1,500: in >= 2 streams %	1,500: core-10 % (range)
SCORE	10.6	54.6	37	19.5 (9.5-34.8)	80.5	45.9 (21.3-68.8)
OFFICIAL	11.6	52.7	11	26.0 (2.4-93.8)	74.0	29.4 (0.0-82.8)
SHOT	15.9	46.8	91	28.4 (21.4-40.9)	71.6	23.3 (13.0-30.7)

slot	full: stream-specific %	full: core-10 %	formula tokens per stream (1,500)	1,500: stream-specific % (range)	1,500: in >= 2 streams %	1,500: core-10 % (range)
JUDGE	17.5	46.3	79	27.6 (22.1-35.7)	72.4	33.9 (24.6-44.6)
STAT	16.8	43.7	25	33.8 (22.0-43.7)	66.2	13.9 (4.2-22.7)
CROWD	23.6	37.0	7	32.5 (4.8-59.7)	67.5	21.8 (0.0-42.8)
OTHER	19.1	44.6	111	30.4 (25.2-36.2)	69.6	28.2 (21.5-37.9)
ALL	17.0	46.2	357	28.4 (24.0-31.9)	71.6	28.7 (25.3-32.6)

Stream-specific share against stream length (descriptive; 2 exploratory p-values): Spearman rho = 0.44 (p = 0.0510) on full texts, -0.66 (p = 0.0015) at 1,500 tokens. Full-text shares depend on how many other long streams exist, matched-size shares on how much of a stream one sample contains.

Table 2.12. Stream-specific share per stream (all slots).

stream	tokens	full: stream-specific %	full: core-10 %	1,500: stream-specific % [2.5-97.5% over subsamples]	1,500: core-10 %
2019 WIM M F Federer-Djokovic	9,791	16.4	43.7	24.0 [14.9, 32.4]	32.4
2023 WIM M F Alcaraz-Djokovic	7,216	15.5	44.8	25.7 [17.9, 34.2]	32.2
2019 AO W F Osaka-Kvitova	6,527	19.0	44.1	26.5 [18.2, 34.5]	25.3
2020 AO M F Djokovic-Thiem	8,371	17.7	43.0	26.8 [17.0, 36.6]	32.6
2020 RG W F Swiatek-Kenin	1,597	11.8	61.6	26.9 [22.3, 31.6]	29.5
2021 AO W F Osaka-Brady	3,276	19.1	49.2	31.9 [26.1, 38.6]	28.7
2021 RG M F Tsitsipas-Djokovic	7,837	20.0	42.7	27.5 [19.3, 36.4]	27.0
2021 USO W F Raducanu-Fernandez	3,498	18.0	48.3	30.6 [22.5, 39.1]	27.8
2022 AO M F Nadal-Medvedev	11,505	19.7	39.8	26.4 [16.6, 36.9]	31.8
2022 RG M QF Djokovic-Nadal	8,046	17.4	43.2	27.2 [18.5, 35.4]	25.5
2022 USO M F Ruud-Alcaraz	6,231	18.5	43.4	30.5 [21.2, 41.7]	26.0
2023 RG W F Swiatek-Muchova	3,800	13.8	51.1	27.3 [20.1, 35.4]	28.5
2023 USO M F Djokovic-Medvedev	5,992	16.4	44.8	30.3 [21.6, 38.8]	25.8
2024 AO M F Sinner-Medvedev	9,256	19.6	42.7	24.0 [15.7, 33.5]	29.2
2024 RG W F Swiatek-Paolini	2,368	16.5	52.4	30.4 [23.7, 36.6]	27.4
2024 WIM W F Paolini-Krejcikova	3,825	17.8	45.2	30.2 [21.9, 38.6]	28.1
2024 USO M F Fritz-Sinner	5,756	16.9	46.2	29.4 [18.6, 39.2]	31.0
2025 AO M SF Sinner-Shelton	7,113	17.8	40.4	30.7 [22.4, 39.7]	28.0
2025 WIM W F Anisimova-Swiatek	2,592	11.9	52.5	31.8 [21.3, 41.6]	30.2
2025 USO M F Sinner-Alcaraz	6,085	16.6	44.7	29.9 [20.7, 40.4]	27.6

2.3 Clusters (exploratory)

Pair similarity = symmetric excess coverage (n >= 2, S = 1,500) or observed Jaccard. Within-minus-between = mean similarity of pairs sharing the label minus the mean of the other pairs; null = labels permuted over the 20 streams (10,000 permutations; the hint cluster enumerated exactly). Mantel rows give the correlation with the pair covariate (node permutation). One-sided p in the direction 'more sharing'.

similarity	label	pairs within	within - between (or r)	null 95%	p
excess_cov_sym	slam	41	0.0065	[-0.0020, 0.0029]	1.0e-04
excess_cov_sym	year	20	-0.0013	[-0.0030, 0.0038]	0.7598
excess_cov_sym	gender	94	0.0014	[-0.0024, 0.0038]	0.1680
excess_cov_sym	hint ('Tim' = 2019 and 2023 finals; other hints single streams)	1	0.0067		0.2316
excess_cov_sym	Mantel r with shared players		0.1461	[-0.1488, 0.1781]	0.0506
excess_cov_sym	Mantel r with	year difference			-0.1037
jaccard_obs	slam	41	0.0112	[-0.0027, 0.0036]	1.0e-04
jaccard_obs	year	20	-0.0027	[-0.0039, 0.0049]	0.8974
jaccard_obs	gender	94	0.0056	[-0.0025, 0.0035]	0.0014
jaccard_obs	hint ('Tim' = 2019 and 2023 finals; other hints single streams)	1	0.0165		0.0579
jaccard_obs	Mantel r with shared players		0.2890	[-0.1418, 0.1562]	0.0008
jaccard_obs	Mantel r with	year difference			0.0238

Table 2.13. QAP regression of pair similarity on same slam, same gender, shared players, |year difference| and same hint (node-permutation p, two-sided; the intercept's p is not meaningful).

similarity	term	beta	p
excess_cov_sym	intercept	0.0751	0.7982
excess_cov_sym	same slam	0.0066	0.0004
excess_cov_sym	same gender	0.0006	0.7208
excess_cov_sym	shared players	0.0029	0.3093
excess_cov_sym		year difference	
excess_cov_sym	same hint (2019-2023 only)	0.0002	0.9963
jaccard_obs	intercept	0.0821	0.0005
jaccard_obs	same slam	0.0110	1.0e-04
jaccard_obs	same gender	0.0042	0.0236
jaccard_obs	shared players	0.0045	0.0693
jaccard_obs		year difference	
jaccard_obs	same hint (2019-2023 only)	0.0018	0.8465

The 2019-2023 pair (both Wimbledon men's finals with Djokovic; hint 'Tim' in both): symmetric excess 8.22 pp, rank 44 of 190 (mean of all pairs 7.56); Jaccard rank 11. The three most similar pairs are 2020 RG W F Swiatek-Kenin / 2023 RG W F Swiatek-Muchova (12.67); 2023 RG W F Swiatek-Muchova / 2024 RG W F Swiatek-Paolini (12.50); 2020 RG W F Swiatek-Kenin / 2024 RG W F Swiatek-Paolini (11.73); all three have same slam, same gender, a shared player (whether they share a broadcaster or commentators is unknown [unverified]).

2.4 Pool -> target coverage for every stream (exploratory, for the record)

I = the other 19 TV streams pooled (raw Phase 2 tokens and 2b normalisation), M = the stream; 95% bootstrap CI over M's utterances (I fixed). Matched I = random 100,000-token subsamples of the other 19 streams (raw tokens; mean and range over 20).

target	raw n>=2	raw n>=3	2b-normalised n>=2	matched I: n>=2 [range]	matched I: n>=3
2019 WIM M F Federer-Djokovic	56.4 [54.6, 58.4]	25.7 [24.0, 27.5]	61.9	55.0 [54.7, 55.3]	24.6
2023 WIM M F Alcaraz-Djokovic	61.0 [59.3, 62.8]	28.9 [26.9, 30.9]	64.2	59.2 [58.8, 59.5]	27.4
2019 AO W F Osaka-Kvitova	53.8 [51.5, 56.2]	26.1 [23.8, 28.3]	59.1	52.2 [51.9, 52.5]	24.6
2020 AO M F Djokovic-Thiem	59.0 [57.3, 60.7]	29.4 [27.3, 31.5]	61.7	57.4 [57.0, 57.8]	28.2
2020 RG W F Swiatek-Kenin	63.4 [59.3, 67.9]	31.3 [27.9, 35.1]	67.1	61.4 [60.8, 62.0]	29.1
2021 AO W F Osaka-Brady	56.3 [53.0, 59.6]	26.9 [24.1, 29.4]	63.2	53.9 [53.0, 54.5]	25.1
2021 RG M F Tsitsipas-Djokovic	58.4 [56.6, 60.1]	28.6 [26.7, 30.5]	62.6	56.7 [56.3, 57.1]	27.2
2021 USO W F Raducanu-Fernandez	52.7 [49.9, 55.6]	23.9 [20.8, 27.0]	57.0	50.6 [50.0, 51.0]	22.3
2022 AO M F Nadal-Medvedev	58.8 [57.1, 60.5]	29.4 [27.7, 31.0]	61.0	57.5 [57.3, 57.7]	28.2
2022 RG M QF Djokovic-Nadal	59.6 [57.9, 61.5]	28.4 [26.5, 30.4]	63.1	58.2 [57.8, 58.5]	26.8
2022 USO M F Ruud-Alcaraz	56.2 [54.0, 58.4]	26.5 [24.4, 28.5]	59.6	54.5 [54.2, 54.9]	25.1
2023 RG W F Swiatek-Muchova	61.3 [58.9, 63.7]	29.6 [26.9, 32.2]	65.3	59.4 [58.8, 59.9]	27.9
2023 USO M F Djokovic-Medvedev	58.6 [56.4, 60.9]	28.3 [26.1, 30.3]	62.2	56.7 [56.4, 57.1]	26.5
2024 AO M F Sinner-Medvedev	56.6 [54.8, 58.4]	28.4 [26.5, 30.0]	59.8	55.1 [54.9, 55.3]	27.0
2024 RG W F Swiatek-Paolini	64.4 [61.8, 67.0]	30.6 [27.6, 33.7]	68.1	62.2 [61.4, 62.7]	28.6
2024 WIM W F Paolini-Krejcikova	56.5 [53.1, 59.5]	28.2 [24.8, 31.4]	60.2	54.4 [53.7, 54.9]	26.2
2024 USO M F Fritz-Sinner	51.1 [48.5, 53.8]	22.7 [20.5, 24.8]	55.5	49.2 [48.9, 49.7]	21.2
2025 AO M SF Sinner-Shelton	57.9 [56.0, 59.8]	29.2 [27.1, 31.3]	62.8	56.2 [55.8, 56.5]	27.6
2025 WIM W F Anisimova-Swiatek	50.2 [46.8, 53.5]	21.9 [18.6, 25.1]	55.3	48.0 [47.2, 48.7]	20.2
2025 USO M F Sinner-Alcaraz	55.5 [53.5, 57.7]	26.0 [23.5, 28.5]	58.8	53.8 [53.3, 54.3]	24.6

2019 ranks 13 of 20 (matched I: 12). With the Phase 2 identification set (18 pool streams) the 2019 value is 55.03%, equal to Phase 2's.

3. Calibration against Parry's definition (2b.2)

Two coders marked formula spans in a fixed sample of 300 utterances (150 from the 2019 final, uniform; 150 from the other 19 streams in proportion to their length; `sample_ids.json`) following `manual.md`. **Both coders are LLM agents, not human specialists**: their agreement measures the reliability of two runs of the same kind of judge, and precision/recall against them measures agreement with that judge, not validity against Parry's concept. Provenance from git: coding_A.jsonl added in 6632cce (2026-10-08 00:26:32 +0000), coding_B.jsonl in 6de4daa (2026-10-08 00:26:12 +0000); co-author trailers 'Claude Fable 5.1 noreply@anthropic.com' / 'Claude Fable 5.1 noreply@anthropic.com'; session identical. Blindness to the automatic inventory rested on the coders' instructions; the Phase 2 report with its formula tables was in the

repository they worked in. A coding of even 50 utterances by a human specialist would be the only check of validity.

Utterances coded by both: 300 (15,829 tokens). Spans in the files: A 1249, B 1010; re-located because `text` did not equal the slice: A 0, B 0; dropped: A 0, B 0.

Table 3.1. Inter-coder agreement (HIGH + MEDIUM spans; bootstrap over utterances, B = 2,000).

token kappa	tokens positive A / B %	spans A / B	span F1 exact	span F1 overlap	matched spans	slot agreement %	type agreement %
0.774 [0.754, 0.796]	25.6 / 21.8	1248 / 1008	0.710 [0.685, 0.736]	0.844 [0.827, 0.861]	952	87.8 (kappa 0.85)	97.6 (kappa 0.95)

LOW-confidence spans: A 1, B 2; including them gives token kappa 0.774, so the plan's 'all confidences' sensitivity is not shown separately. Span counts by slot: A CROWD 43, JUDGE 392, OFFICIAL 62, OTHER 52, SCORE 275, SHOT 260, STAT 165; B CROWD 34, JUDGE 267, OFFICIAL 53, OTHER 43, SCORE 219, SHOT 227, STAT 167.

Table 3.2. Automatic measures against the coders, with chance levels (token level, HIGH + MEDIUM spans; 'Pool' = the 18 pool streams minus the utterance's own; 'in-sample' = the utterance's own stream; bootstrap 95% CI over utterances). Chance precision = the reference's positive share; chance recall = the measure's positive share; ceiling = the highest precision possible at that positive share. The last three measures (marked exploratory) were added in the plan beyond the brief.

automatic measure	reference	auto-positive %	precision	chance precision	ceiling	precision - chance	recall	chance recall	kappa	F1
pool exact n >= 2 (Phase 2 (a))	A	53.7	0.33 [0.31, 0.35]	0.26	0.48	0.07 [0.06, 0.08]	0.69 [0.67, 0.71]	0.54	0.147 [0.128, 0.169]	0.44
pool exact n >= 2 (Phase 2 (a))	B	53.7	0.29 [0.27, 0.31]	0.22	0.41	0.07 [0.06, 0.08]	0.71 [0.69, 0.74]	0.54	0.149 [0.130, 0.169]	0.41
pool exact n >= 2 (Phase 2 (a))	A and B	53.7	0.26 [0.24, 0.29]	0.20	0.37	0.07 [0.06, 0.08]	0.72 [0.70, 0.74]	0.54	0.137 [0.119, 0.156]	0.39
pool exact n >= 2 (Phase 2 (a))	A or B	53.7	0.35 [0.33, 0.38]	0.28	0.52	0.08 [0.07, 0.09]	0.68 [0.67, 0.70]	0.54	0.159 [0.140, 0.181]	0.47
pool exact n >= 3	A	25.0	0.36 [0.33, 0.39]	0.26	1.00	0.10 [0.08, 0.12]	0.35 [0.32, 0.38]	0.25	0.134 [0.108, 0.162]	0.35
pool exact n >= 3	B	25.0	0.33 [0.30, 0.36]	0.22	0.87	0.11 [0.09, 0.13]	0.38 [0.35, 0.40]	0.25	0.155 [0.129, 0.183]	0.35
pool exact n >= 3	A and B	25.0	0.30 [0.27, 0.32]	0.20	0.78	0.10 [0.08, 0.12]	0.38 [0.35, 0.41]	0.25	0.143 [0.116, 0.171]	0.33
pool exact n >= 3	A or B	25.0	0.39 [0.36, 0.42]	0.28	1.00	0.11 [0.09, 0.14]	0.35 [0.33, 0.38]	0.25	0.146 [0.121, 0.173]	0.37
pool systems only ((b))	A	37.9	0.37 [0.35, 0.40]	0.26	0.68	0.12 [0.10, 0.13]	0.55 [0.53, 0.58]	0.38	0.203 [0.179, 0.229]	0.45
pool systems only ((b))	B	37.9	0.34 [0.31, 0.37]	0.22	0.58	0.12 [0.11, 0.14]	0.59 [0.56, 0.62]	0.38	0.212 [0.186, 0.238]	0.43
pool systems only ((b))	A and B	37.9	0.31 [0.28, 0.33]	0.20	0.52	0.11 [0.10, 0.13]	0.59 [0.56, 0.62]	0.38	0.197 [0.174, 0.223]	0.40
pool systems only ((b))	A or B	37.9	0.41 [0.38, 0.44]	0.28	0.73	0.13 [0.11, 0.14]	0.55 [0.53, 0.58]	0.38	0.217 [0.191, 0.244]	0.47

automatic measure	reference	auto-positive %	precision	chance precision	ceiling	precision - chance	recall	chance recall	kappa	F1
pool (a) or (b)	A	62.3	0.32 [0.30, 0.34]	0.26	0.41	0.06 [0.06, 0.07]	0.78 [0.76, 0.80]	0.62	0.144 [0.126, 0.165]	0.45
pool (a) or (b)	B	62.3	0.28 [0.26, 0.31]	0.22	0.35	0.07 [0.06, 0.07]	0.81 [0.79, 0.83]	0.62	0.143 [0.126, 0.163]	0.42
pool (a) or (b)	A and B	62.3	0.26 [0.24, 0.28]	0.20	0.32	0.06 [0.05, 0.07]	0.81 [0.79, 0.83]	0.62	0.130 [0.113, 0.148]	0.39
pool (a) or (b)	A or B	62.3	0.35 [0.32, 0.37]	0.28	0.45	0.07 [0.06, 0.08]	0.78 [0.76, 0.80]	0.62	0.158 [0.139, 0.180]	0.48
in-sample exact n >= 2	A	40.4	0.36 [0.33, 0.38]	0.26	0.63	0.10 [0.09, 0.11]	0.56 [0.54, 0.59]	0.40	0.178 [0.155, 0.201]	0.44
in-sample exact n >= 2	B	40.4	0.32 [0.30, 0.35]	0.22	0.54	0.11 [0.09, 0.12]	0.60 [0.58, 0.63]	0.40	0.192 [0.168, 0.217]	0.42
in-sample exact n >= 2	A and B	40.4	0.30 [0.27, 0.32]	0.20	0.49	0.10 [0.09, 0.11]	0.61 [0.58, 0.63]	0.40	0.182 [0.159, 0.205]	0.40
in-sample exact n >= 2	A or B	40.4	0.39 [0.36, 0.41]	0.28	0.69	0.11 [0.09, 0.12]	0.56 [0.53, 0.58]	0.40	0.189 [0.164, 0.214]	0.46
in-sample exact n >= 3	A	18.4	0.40 [0.36, 0.43]	0.26	1.00	0.14 [0.11, 0.17]	0.29 [0.26, 0.31]	0.18	0.151 [0.120, 0.180]	0.33
in-sample exact n >= 3	B	18.4	0.38 [0.35, 0.42]	0.22	1.00	0.16 [0.13, 0.19]	0.32 [0.29, 0.35]	0.18	0.184 [0.153, 0.217]	0.35
in-sample exact n >= 3	A and B	18.4	0.34 [0.31, 0.38]	0.20	1.00	0.15 [0.12, 0.17]	0.32 [0.29, 0.35]	0.18	0.174 [0.142, 0.207]	0.33
in-sample exact n >= 3	A or B	18.4	0.44 [0.40, 0.47]	0.28	1.00	0.16 [0.13, 0.19]	0.29 [0.26, 0.31]	0.18	0.161 [0.132, 0.192]	0.35
pool exact n >= 2, 2b normalisation (exploratory)	A	58.1	0.32 [0.30, 0.35]	0.26	0.44	0.07 [0.06, 0.08]	0.73 [0.71, 0.75]	0.58	0.143 [0.124, 0.165]	0.45
pool exact n >= 2, 2b normalisation (exploratory)	B	58.1	0.29 [0.27, 0.31]	0.22	0.38	0.07 [0.06, 0.08]	0.77 [0.75, 0.79]	0.58	0.148 [0.130, 0.168]	0.42
pool exact n >= 2, 2b normalisation (exploratory)	A and B	58.1	0.26 [0.24, 0.28]	0.20	0.34	0.06 [0.06, 0.07]	0.77 [0.75, 0.79]	0.58	0.135 [0.118, 0.154]	0.39
pool exact n >= 2, 2b normalisation (exploratory)	A or B	58.1	0.35 [0.33, 0.37]	0.28	0.48	0.07 [0.06, 0.08]	0.73 [0.71, 0.75]	0.58	0.156 [0.136, 0.178]	0.47
pool content n >= 2 (exploratory)	A	51.1	0.32 [0.29, 0.34]	0.26	0.50	0.06 [0.05, 0.07]	0.63 [0.61, 0.65]	0.51	0.121 [0.101, 0.142]	0.42
pool content n >= 2 (exploratory)	B	51.1	0.28 [0.26, 0.30]	0.22	0.43	0.06 [0.05, 0.07]	0.65 [0.63, 0.68]	0.51	0.122 [0.103, 0.143]	0.39
pool content n >= 2 (exploratory)	A and B	51.1	0.25 [0.23, 0.27]	0.20	0.38	0.06 [0.05, 0.07]	0.66 [0.63, 0.68]	0.51	0.111 [0.092, 0.131]	0.36
pool content n >= 2 (exploratory)	A or B	51.1	0.34 [0.32, 0.37]	0.28	0.54	0.07 [0.05, 0.08]	0.63 [0.61, 0.65]	0.51	0.132 [0.111, 0.154]	0.44

automatic measure	reference	auto-positive %	precision	chance precision	ceiling	precision - chance	recall	chance recall	kappa	F1
genre core: in >= 5 other streams' inventories (exploratory)	A	26.7	0.36 [0.33, 0.39]	0.26	0.96	0.11 [0.09, 0.13]	0.38 [0.35, 0.40]	0.27	0.147 [0.121, 0.173]	0.37
genre core: in >= 5 other streams' inventories (exploratory)	B	26.7	0.34 [0.31, 0.37]	0.22	0.82	0.12 [0.10, 0.14]	0.41 [0.39, 0.44]	0.27	0.173 [0.147, 0.198]	0.37
genre core: in >= 5 other streams' inventories (exploratory)	A and B	26.7	0.31 [0.28, 0.34]	0.20	0.74	0.11 [0.10, 0.13]	0.42 [0.39, 0.45]	0.27	0.168 [0.144, 0.194]	0.36
genre core: in >= 5 other streams' inventories (exploratory)	A or B	26.7	0.39 [0.36, 0.42]	0.28	1.00	0.11 [0.09, 0.13]	0.37 [0.35, 0.40]	0.27	0.151 [0.124, 0.177]	0.38

Table 3.2b. By sample group (pool n >= 2, pool n >= 3 and pool systems against coder A, coder B and their union). The two halves differ in design: 150 uniform 2019 utterances against 150 token-weighted utterances of the other streams, identified against a pool of 17 instead of 18 streams.

measure	reference	group	utterances	median tokens	auto-positive %	precision	chance precision	precision - chance	recall	chance recall	ka
pool exact n >= 2 (Phase 2 (a))	A	2019	150	26	54.1	0.43	0.34	0.09	0.68	0.54	0.
pool exact n >= 2 (Phase 2 (a))	A	other	150	59	53.5	0.27	0.21	0.06	0.69	0.54	0.
pool exact n >= 2 (Phase 2 (a))	B	2019	150	26	54.1	0.41	0.31	0.10	0.71	0.54	0.
pool exact n >= 2 (Phase 2 (a))	B	other	150	59	53.5	0.23	0.17	0.06	0.72	0.54	0.
pool exact n >= 2 (Phase 2 (a))	A or B	2019	150	26	54.1	0.47	0.38	0.10	0.68	0.54	0.
pool exact n >= 2 (Phase 2 (a))	A or B	other	150	59	53.5	0.29	0.23	0.07	0.69	0.54	0.
pool exact n >= 3	A	2019	150	26	24.4	0.48	0.34	0.14	0.34	0.24	0.
pool exact n >= 3	A	other	150	59	25.4	0.30	0.21	0.09	0.36	0.25	0.

measure	reference	group	utterances	median tokens	auto-positive %	precision	chance precision	precision - chance	recall	chance recall	ka
pool exact n >= 3	B	2019	150	26	24.4	0.46	0.31	0.15	0.36	0.24	0.1
pool exact n >= 3	B	other	150	59	25.4	0.27	0.17	0.10	0.40	0.25	0.1
pool exact n >= 3	A or B	2019	150	26	24.4	0.52	0.38	0.15	0.34	0.24	0.1
pool exact n >= 3	A or B	other	150	59	25.4	0.33	0.23	0.10	0.36	0.25	0.1
pool systems only ((b))	A	2019	150	26	40.1	0.49	0.34	0.15	0.57	0.40	0.2
pool systems only ((b))	A	other	150	59	36.8	0.31	0.21	0.10	0.54	0.37	0.1
pool systems only ((b))	B	2019	150	26	40.1	0.46	0.31	0.15	0.60	0.40	0.2
pool systems only ((b))	B	other	150	59	36.8	0.27	0.17	0.10	0.58	0.37	0.1
pool systems only ((b))	A or B	2019	150	26	40.1	0.54	0.38	0.16	0.57	0.40	0.2
pool systems only ((b))	A or B	other	150	59	36.8	0.33	0.23	0.10	0.53	0.37	0.1

Table 3.2c. Span level (share of each coder's HIGH spans lying wholly inside, or touching, the automatic positive set; chance = mean over 200 random circular shifts of the automatic labels within each utterance, with the 97.5th percentile).

measure	coder	HIGH spans	wholly inside	chance (97.5%)	touching	chance (97.5%)
pool exact n >= 2 (Phase 2 (a))	A	577	63.8	34.5 (37.8)	89.1	78.6 (81.5)
pool exact n >= 2 (Phase 2 (a))	B	428	61.9	33.6 (37.6)	89.0	79.0 (82.7)
pool exact n >= 3	A	577	31.5	14.1 (16.8)	45.8	39.5 (43.2)
pool exact n >= 3	B	428	26.6	12.9 (15.7)	44.6	39.9 (43.5)
pool systems only ((b))	A	577	58.4	29.0 (31.7)	77.3	60.1 (63.4)
pool systems only ((b))	B	428	56.8	27.5 (30.6)	78.3	60.7 (64.5)
pool (a) or (b)	A	577	79.5	48.0 (51.3)	94.3	84.8 (87.4)
pool (a) or (b)	B	428	79.0	46.7 (50.7)	94.6	85.0 (88.1)
in-sample exact n >= 2	A	577	53.9	24.2 (26.7)	78.2	66.1 (69.2)
in-sample exact n >= 2	B	428	51.6	23.7 (27.3)	79.9	67.1 (71.3)
in-sample exact n >= 3	A	577	27.4	10.5 (12.7)	38.0	31.4 (34.8)
in-sample exact n >= 3	B	428	21.0	9.6 (12.1)	34.8	32.1 (35.8)
pool exact n >= 2, 2b normalisation (exploratory)	A	577	73.3	42.1 (45.2)	93.4	83.1 (85.5)
pool exact n >= 2, 2b normalisation (exploratory)	B	428	73.8	41.6 (45.6)	94.2	83.6 (86.7)
pool content n >= 2 (exploratory)	A	577	52.0	30.0 (33.4)	77.8	73.7 (77.0)
pool content n >= 2 (exploratory)	B	428	46.5	28.8 (32.7)	74.8	73.8 (77.3)
genre core: in >= 5 other streams' inventories (exploratory)	A	577	36.2	13.2 (15.6)	65.9	50.2 (53.7)
genre core: in >= 5 other streams' inventories (exploratory)	B	428	37.9	12.7 (15.4)	69.2	51.3 (55.4)

Table 3.3. Strata (pool n >= 2 and pool n >= 3 against each coder): recall by the coder's slot and by span confidence (HIGH, MEDIUM; LOW has 1-2 spans per coder and is omitted); precision by the classifier's token slot (span mode on single tokens).

measure	coder	kind	stratum	numerator	denominator	value
pool_n2	A	recall by coder slot	CROWD	80	175	0.457
pool_n2	A	recall by coder slot	JUDGE	825	1392	0.593
pool_n2	A	recall by coder slot	OFFICIAL	172	216	0.796
pool_n2	A	recall by coder slot	OTHER	108	153	0.706
pool_n2	A	recall by coder slot	SCORE	669	820	0.816
pool_n2	A	recall by coder slot	SHOT	528	713	0.741
pool_n2	A	recall by coder slot	STAT	402	588	0.684
pool_n2	A	share of coder tokens covered, by confidence	HIGH	1378	1735	0.794
pool_n2	A	share of coder tokens covered, by confidence	MEDIUM	1406	2322	0.606
pool_n2	A	precision by classifier slot	CROWD	53	161	0.329
pool_n2	A	precision by classifier slot	JUDGE	406	1870	0.217
pool_n2	A	precision by classifier slot	OFFICIAL	162	255	0.635
pool_n2	A	precision by classifier slot	OTHER	721	3079	0.234
pool_n2	A	precision by classifier slot	SCORE	422	634	0.666
pool_n2	A	precision by classifier slot	SHOT	745	1848	0.403
pool_n2	A	precision by classifier slot	STAT	275	659	0.417
pool_n2	B	recall by coder slot	CROWD	62	132	0.470
pool_n2	B	recall by coder slot	JUDGE	635	991	0.641
pool_n2	B	recall by coder slot	OFFICIAL	173	213	0.812
pool_n2	B	recall by coder slot	OTHER	103	141	0.730
pool_n2	B	recall by coder slot	SCORE	542	673	0.805
pool_n2	B	recall by coder slot	SHOT	515	676	0.762
pool_n2	B	recall by coder slot	STAT	440	630	0.698
pool_n2	B	share of coder tokens covered, by confidence	HIGH	1084	1374	0.789
pool_n2	B	share of coder tokens covered, by confidence	MEDIUM	1386	2082	0.666
pool_n2	B	precision by classifier slot	CROWD	46	161	0.286
pool_n2	B	precision by classifier slot	JUDGE	300	1870	0.160
pool_n2	B	precision by classifier slot	OFFICIAL	166	255	0.651
pool_n2	B	precision by classifier slot	OTHER	609	3079	0.198
pool_n2	B	precision by classifier slot	SCORE	423	634	0.667
pool_n2	B	precision by classifier slot	SHOT	679	1848	0.367
pool_n2	B	precision by classifier slot	STAT	247	659	0.375
pool_n3	A	recall by coder slot	CROWD	31	175	0.177
pool_n3	A	recall by coder slot	JUDGE	378	1392	0.272
pool_n3	A	recall by coder slot	OFFICIAL	92	216	0.426
pool_n3	A	recall by coder slot	OTHER	40	153	0.261
pool_n3	A	recall by coder slot	SCORE	375	820	0.457
pool_n3	A	recall by coder slot	SHOT	295	713	0.414
pool_n3	A	recall by coder slot	STAT	206	588	0.350
pool_n3	A	share of coder tokens covered, by confidence	HIGH	785	1735	0.452
pool_n3	A	share of coder tokens covered, by confidence	MEDIUM	632	2322	0.272
pool_n3	A	precision by classifier slot	CROWD	12	48	0.250
pool_n3	A	precision by classifier slot	JUDGE	204	917	0.222
pool_n3	A	precision by classifier slot	OFFICIAL	96	126	0.762
pool_n3	A	precision by classifier slot	OTHER	348	1389	0.251
pool_n3	A	precision by classifier slot	SCORE	195	304	0.641
pool_n3	A	precision by classifier slot	SHOT	391	852	0.459

measure	coder	kind	stratum	numerator	denominator	value
pool_n3	A	precision by classifier slot	STAT	171	326	0.525
pool_n3	B	recall by coder slot	CROWD	16	132	0.121
pool_n3	B	recall by coder slot	JUDGE	335	991	0.338
pool_n3	B	recall by coder slot	OFFICIAL	97	213	0.455
pool_n3	B	recall by coder slot	OTHER	44	141	0.312
pool_n3	B	recall by coder slot	SCORE	273	673	0.406
pool_n3	B	recall by coder slot	SHOT	300	676	0.444
pool_n3	B	recall by coder slot	STAT	241	630	0.383
pool_n3	B	share of coder tokens covered, by confidence	HIGH	606	1374	0.441
pool_n3	B	share of coder tokens covered, by confidence	MEDIUM	700	2082	0.336
pool_n3	B	precision by classifier slot	CROWD	5	48	0.104
pool_n3	B	precision by classifier slot	JUDGE	159	917	0.173
pool_n3	B	precision by classifier slot	OFFICIAL	100	126	0.794
pool_n3	B	precision by classifier slot	OTHER	312	1389	0.225
pool_n3	B	precision by classifier slot	SCORE	208	304	0.684
pool_n3	B	precision by classifier slot	SHOT	366	852	0.430
pool_n3	B	precision by classifier slot	STAT	156	326	0.479

Table 3.4. Slot classifier against the coders' slot labels (rows = coder, columns = classifier). Span mode on every coder span (the mode used for formula spans), context mode on every coder span, and context and span mode on the referring expressions of the sample that lie inside a coder span (the unit and mode of H4). The context-mode row on coder spans compares the classifier with labels the coders assigned from the very words that context mode excludes, so it describes the design rather than the classifier's error on references; the reference rows are the relevant ones (their bootstrap intervals are in the Key results).

unit and mode	coder A: agreement % / kappa / n	coder B: agreement % / kappa / n
coder spans, span mode	57.9 / 0.490 / 1249	57.3 / 0.484 / 1010
coder spans, context mode	25.9 / 0.141 / 1249	25.7 / 0.146 / 1010
commentary references, context mode (H4)	46.2 / 0.354 / 65	49.3 / 0.388 / 69
commentary references, span mode	46.2 / 0.343 / 65	47.8 / 0.364 / 69
all references, context mode	60.9 / 0.541 / 92	62.9 / 0.561 / 97

Coder A, span mode, coder spans:

coder A \ classifier	SCORE	OFFICIAL	SHOT	JUDGE	STAT	CROWD	OTHER	total
CROWD	3	0	1	6	3	20	10	43
JUDGE	16	4	82	170	9	1	110	392
OFFICIAL	1	53	4	1	0	0	3	62
OTHER	3	2	11	9	2	0	25	52
SCORE	155	0	20	18	11	0	71	275
SHOT	2	0	216	15	2	1	24	260
STAT	6	1	36	5	84	2	31	165

Coder A, context mode, commentary references:

coder A \ classifier	SCORE	OFFICIAL	SHOT	JUDGE	STAT	CROWD	OTHER	total
CROWD	1	0	0	0	0	1	2	4
JUDGE	2	1	4	6	0	0	10	23
OTHER	0	0	2	0	0	0	5	7
SCORE	3	0	1	0	1	0	0	5
SHOT	0	0	9	2	1	0	1	13
STAT	0	0	0	0	6	0	7	13

Coder B, span mode, coder spans:

coder B \ classifier	SCORE	OFFICIAL	SHOT	JUDGE	STAT	CROWD	OTHER	total
CROWD	2	0	1	2	1	15	13	34
JUDGE	16	0	55	111	10	1	74	267
OFFICIAL	1	46	4	0	0	0	2	53
OTHER	1	3	17	6	1	1	14	43
SCORE	143	0	15	8	7	0	46	219
SHOT	1	1	183	13	9	1	19	227
STAT	12	0	34	10	67	2	42	167

Coder B, context mode, commentary references:

coder B \ classifier	SCORE	OFFICIAL	SHOT	JUDGE	STAT	CROWD	OTHER	total
CROWD	0	0	0	0	0	3	2	5
JUDGE	2	0	4	9	1	0	9	25
OTHER	1	0	2	0	1	0	5	9
SCORE	4	0	2	1	1	0	1	9
SHOT	0	0	8	1	0	0	1	10
STAT	0	0	0	0	5	0	6	11

4. Thrift and extension per situational slot (2b.3)

Referring expressions were extracted for the two players of every TV stream (2998 references; 123 inside umpire patterns; 42 descriptive epithets without a resolved referent). The extraction reproduces Phase 2's totals for the two finals (tv_2019wimF: djokovic 188, federer 164, tv_2023wimF: alcaraz 80, djokovic 121). The situational slot of a reference is the context-mode slot (section 1; its agreement with the coders is in Table 3.4).

Table 4.1. How each stream names the players (commentary-only resolved references; shares %; epithet upper bound counts every unresolved epithet as a player reference).

stream	references	surname	first name	full name	hypocoristic	epithet	epithet (upper)
2019 WIM M F Federer-Djokovic	321	73	14	8	0	5	5
2023 WIM M F Alcaraz-Djokovic	170	78	14	4	0	5	5
2019 AO W F Osaka-Kvitova	173	66	17	16	0	1	3
2020 AO M F Djokovic-Thiem	164	39	41	16	0	3	3
2020 RG W F Swiatek-Kenin	41	98	0	2	0	0	0
2021 AO W F Osaka-Brady	132	93	1	6	0	0	0
2021 RG M F Tsitsipas-Djokovic	233	91	1	5	0	3	8
2021 USO W F Raducanu-Fernandez	55	84	7	9	0	0	4
2022 AO M F Nadal-Medvedev	232	54	2	3	34	7	7
2022 RG M QF Djokovic-Nadal	288	77	2	2	19	0	1
2022 USO M F Ruud-Alcaraz	80	79	6	12	0	2	6
2023 RG W F Swiatek-Muchova	109	96	0	1	0	3	7
2023 USO M F Djokovic-Medvedev	140	87	11	1	0	0	0

stream	references	surname	first name	full name	hypocoristic	epithet	epithet (upper)
2024 AO M F Sinner-Medvedev	192	78	16	4	0	3	3
2024 RG W F Swiatek-Paolini	65	98	0	2	0	0	7
2024 WIM W F Paolini-Krejcikova	65	74	20	5	0	2	2
2024 USO M F Fritz-Sinner	104	33	40	21	0	6	7
2025 AO M SF Sinner-Shelton	119	52	25	18	0	4	7
2025 WIM W F Anisimova-Swiatek	39	56	21	21	0	3	5
2025 USO M F Sinner-Alcaraz	111	69	23	4	0	4	5
all 20 TV streams	2833	73	12	7	5	3	4

Table 4.2. Economy of naming by situational slot (commentary-only references, all streams; per stream x player x slot cell: modal-form share and distinct forms per 100 references, averaged over cells with >= 5 references; distinct forms per 100 tokens of the slot, averaged over cells).

slot	references	cells	cells >= 5	modal share %	distinct per 100 refs	distinct per 100 slot tokens
SCORE	194	37	14	83.1	27.0	0.53
OFFICIAL	37	16	3	86.7	25.6	1.48
SHOT	618	40	36	80.4	16.8	0.19
JUDGE	535	40	32	77.2	17.5	0.21
STAT	142	36	12	74.4	31.7	0.45
CROWD	55	23	1	87.5	25.0	1.03
OTHER	1252	40	40	73.4	13.4	0.14

Table 4.3. One form per slot? (cells = player x slot with >= 5 commentary references; criterion in plan section 9).

stream	slot type	cells >= 5	mean modal share	distinct forms per 100 refs	cells with modal share >= 0.9	lowest modal share	verdict
2019 WIM M F Federer-Djokovic	situational	11	0.73	19.1	1	0.44	not one form per slot
2023 WIM M F Alcaraz-Djokovic	situational	9	0.77	20.3	1	0.50	not one form per slot
2019 AO W F Osaka-Kvitova	situational	6	0.63	14.1	0	0.53	not one form per slot
2020 AO M F Djokovic-Thiem	situational	6	0.53	17.2	0	0.36	not one form per slot
2020 RG W F Swiatek-Kenin	situational	4	1.00	16.7	4	1.00	one form throughout (economy without slot conditioning)
2021 AO W F Osaka-Brady	situational	8	0.96	10.9	6	0.83	not one form per slot
2021 RG M F Tsitsipas-Djokovic	situational	11	0.91	12.8	7	0.67	not one form per slot
2021 USO W F Raducanu-Fernandez	situational	4	0.86	17.1	2	0.60	not one form per slot
2022 AO M F Nadal-Medvedev	situational	8	0.76	17.8	0	0.56	not one form per slot

stream	slot type	cells >= 5	mean modal share	distinct forms per 100 refs	cells with modal share >= 0.9	lowest modal share	verdict
2022 RG M QF Djokovic-Nadal	situational	11	0.81	16.2	4	0.61	not one form per slot
2022 USO M F Ruud-Alcaraz	situational	5	0.80	20.4	2	0.30	not one form per slot
2023 RG W F Swiatek-Muchova	situational	6	0.96	9.1	5	0.87	not one form per slot
2023 USO M F Djokovic-Medvedev	situational	9	0.88	18.6	4	0.70	not one form per slot
2024 AO M F Sinner-Medvedev	situational	6	0.74	12.2	2	0.44	not one form per slot
2024 RG W F Swiatek-Paolini	situational	5	0.99	11.9	5	0.94	one form throughout (economy without slot conditioning)
2024 WIM W F Paolini-Krejcikova	situational	5	0.74	24.9	1	0.50	not one form per slot
2024 USO M F Fritz-Sinner	situational	6	0.53	29.4	0	0.41	not one form per slot
2025 AO M SF Sinner-Shelton	situational	8	0.59	25.6	0	0.33	not one form per slot
2025 WIM W F Anisimova-Swiatek	situational	3	0.60	37.1	0	0.43	not one form per slot
2025 USO M F Sinner-Alcaraz	situational	7	0.66	31.1	0	0.33	not one form per slot
2019 WIM M F Federer-Djokovic	syntactic	12	0.74	20.6	2	0.48	not one form per slot
2023 WIM M F Alcaraz-Djokovic	syntactic	8	0.79	20.1	1	0.62	not one form per slot
2019 AO W F Osaka-Kvitova	syntactic	11	0.67	22.4	2	0.42	not one form per slot
2020 AO M F Djokovic-Thiem	syntactic	9	0.56	31.8	0	0.40	not one form per slot
2020 RG W F Swiatek-Kenin	syntactic	4	0.97	16.9	3	0.89	not one form per slot
2021 AO W F Osaka-Brady	syntactic	8	0.88	18.3	5	0.60	not one form per slot
2021 RG M F Tsitsipas-Djokovic	syntactic	11	0.88	16.1	6	0.67	not one form per slot
2021 USO W F Raducanu-Fernandez	syntactic	5	0.92	22.0	3	0.80	not one form per slot
2022 AO M F Nadal-Medvedev	syntactic	12	0.78	20.0	3	0.57	not one form per slot
2022 RG M QF Djokovic-Nadal	syntactic	12	0.78	14.4	4	0.55	not one form per slot
2022 USO M F Ruud-Alcaraz	syntactic	5	0.81	18.7	2	0.50	not one form per slot
2023 RG W F Swiatek-Muchova	syntactic	7	0.98	12.2	6	0.83	not one form per slot
2023 USO M F Djokovic-Medvedev	syntactic	10	0.88	16.9	6	0.50	not one form per slot
2024 AO M F Sinner-	syntactic	10	0.78	18.5	3	0.39	not one form per slot

stream	slot type	cells >= 5	mean modal share	distinct forms per 100 refs	cells with modal share >= 0.9	lowest modal share	verdict
Medvedev							
2024 RG W F Swiatek-Paolini	syntactic	6	0.99	15.0	6	0.92	one form throughout (economy without slot conditioning)
2024 WIM W F Paolini-Krejcikova	syntactic	6	0.84	25.7	2	0.56	not one form per slot
2024 USO M F Fritz-Sinner	syntactic	7	0.61	27.9	1	0.33	not one form per slot
2025 AO M SF Sinner-Shelton	syntactic	9	0.57	30.3	0	0.35	not one form per slot
2025 WIM W F Anisimova-Swiatek	syntactic	3	0.66	35.9	0	0.55	not one form per slot
2025 USO M F Sinner-Alcaraz	syntactic	7	0.72	27.4	0	0.50	not one form per slot

Table 4.4. Thrift tests as pre-registered (D = sum over cells of distinct categories (H4a) or forms (H4b); one-sided p for fewer than under the null; 10,000 permutations).

test	references	slot type	stream	tokens	D	null mean	null 95%	p	status
H4a	commentary_only	situational		2833	303	388.4	[377, 399]	1.0e-04	2b-primary
H4b	commentary_only	situational		2833	472	483.3	[473, 494]	0.0221	2b-primary
H4a	all_references	situational		2956	310	402.7	[391, 414]	1.0e-04	exploratory (sensitivity)
H4b	all_references	situational		2956	486	510.1	[499, 521]	1.0e-04	exploratory (sensitivity)
H4a	commentary_only	syntactic		2833	297	387.8	[377, 398]	1.0e-04	exploratory
H4b	commentary_only	syntactic		2833	474	495.7	[485, 506.025]	1.0e-04	exploratory
H4b per team	commentary_only	slot_sit	2019 WIM M F Federer-Djokovic	321	48	47.9		0.6295	exploratory
H4b per team	commentary_only	slot_sit	2023 WIM M F Alcaraz-Djokovic	170	32	33.6		0.2506	exploratory
H4b per team	commentary_only	slot_sit	2019 AO W F Osaka-Kvitova	173	28	28.9		0.3820	exploratory
H4b per team	commentary_only	slot_sit	2020 AO M F Djokovic-Thiem	164	26	28.3		0.0931	exploratory
H4b per team	commentary_only	slot_sit	2020 RG W F Swiatek-Kenin	41	11	11.0		1.0000	exploratory
H4b per team	commentary_only	slot_sit	2021 AO W F Osaka-Brady	132	17	16.5		0.9294	exploratory
H4b per team	commentary_only	slot_sit	2021 RG M F Tsitsipas-Djokovic	233	24	24.7		0.4384	exploratory
H4b per team	commentary_only	slot_sit	2021 USO W F Raducanu-Fernandez	55	14	17.4		0.0008	exploratory
H4b per	commentary_only	slot_sit	2022 AO M F Nadal-	232	32	31.9		0.6315	exploratory

test	references	slot type	stream	tokens	D	null mean	null 95%	p	status
team			Medvedev						
H4b per team	commentary_only	slot_sit	2022 RG M QF Djokovic-Nadal	288	25	27.5		0.0286	exploratory
H4b per team	commentary_only	slot_sit	2022 USO M F Ruud-Alcaraz	80	21	21.4		0.5207	exploratory
H4b per team	commentary_only	slot_sit	2023 RG W F Swiatek-Muchova	109	14	15.0		0.2020	exploratory
H4b per team	commentary_only	slot_sit	2023 USO M F Djokovic-Medvedev	140	19	19.1		0.6202	exploratory
H4b per team	commentary_only	slot_sit	2024 AO M F Sinner-Medvedev	192	26	27.8		0.1737	exploratory
H4b per team	commentary_only	slot_sit	2024 RG W F Swiatek-Paolini	65	10	10.0		1.0000	exploratory
H4b per team	commentary_only	slot_sit	2024 WIM W F Paolini-Krejcikova	65	19	19.5		0.4935	exploratory
H4b per team	commentary_only	slot_sit	2024 USO M F Fritz-Sinner	104	31	29.7		0.8958	exploratory
H4b per team	commentary_only	slot_sit	2025 AO M SF Sinner-Shelton	119	28	28.4		0.5071	exploratory
H4b per team	commentary_only	slot_sit	2025 WIM W F Anisimova-Swiatek	39	19	17.4		0.9821	exploratory
H4b per team	commentary_only	slot_sit	2025 USO M F Sinner-Alcaraz	111	28	27.2		0.8527	exploratory
H4b per team	commentary_only	slot_syn	2019 WIM M F Federer-Djokovic	321	45	45.8		0.4220	exploratory
H4b per team	commentary_only	slot_syn	2023 WIM M F Alcaraz-Djokovic	170	28	29.2		0.3131	exploratory
H4b per team	commentary_only	slot_syn	2019 AO W F Osaka-Kvitova	173	30	33.4		0.0302	exploratory
H4b per team	commentary_only	slot_syn	2020 AO M F Djokovic-Thiem	164	32	32.3		0.5563	exploratory
H4b per team	commentary_only	slot_syn	2020 RG W F Swiatek-Kenin	41	11	10.9		1.0000	exploratory
H4b per team	commentary_only	slot_syn	2021 AO W F Osaka-Brady	132	18	17.0		0.9736	exploratory
H4b per team	commentary_only	slot_syn	2021 RG M F Tsitsipas-Djokovic	233	23	22.8		0.6974	exploratory
H4b per team	commentary_only	slot_syn	2021 USO W F Raducanu-Fernandez	55	19	19.6		0.4338	exploratory
H4b per	commentary_only	slot_syn	2022 AO M F Nadal-	232	31	32.7		0.2372	exploratory

test	references	slot type	stream	tokens	D	null mean	null 95%	p	status
team			Medvedev						
H4b per team	commentary_only	slot_syn	2022 RG M QF Djokovic-Nadal	288	27	26.6		0.7986	exploratory
H4b per team	commentary_only	slot_syn	2022 USO M F Ruud-Alcaraz	80	19	23.0		0.0079	exploratory
H4b per team	commentary_only	slot_syn	2023 RG W F Swiatek-Muchova	109	13	13.9		0.2518	exploratory
H4b per team	commentary_only	slot_syn	2023 USO M F Djokovic-Medvedev	140	18	18.8		0.3641	exploratory
H4b per team	commentary_only	slot_syn	2024 AO M F Sinner-Medvedev	192	27	29.4		0.0941	exploratory
H4b per team	commentary_only	slot_syn	2024 RG W F Swiatek-Paolini	65	13	12.9		1.0000	exploratory
H4b per team	commentary_only	slot_syn	2024 WIM W F Paolini-Krejcikova	65	19	20.3		0.2252	exploratory
H4b per team	commentary_only	slot_syn	2024 USO M F Fritz-Sinner	104	25	28.9		0.0070	exploratory
H4b per team	commentary_only	slot_syn	2025 AO M SF Sinner-Shelton	119	30	32.7		0.0558	exploratory
H4b per team	commentary_only	slot_syn	2025 WIM W F Anisimova-Swiatek	39	19	18.8		0.7276	exploratory
H4b per team	commentary_only	slot_syn	2025 USO M F Sinner-Alcaraz	111	27	26.4		0.8067	exploratory

Table 4.4b. POST HOC (addendum 1, M5): robustness of H4a and H4b (one-sided p; Monte Carlo SE; Holm p when the row's p replaces the pre-registered p in the seven-test family).

test	run	slot	tokens	streams	D	null mean	p	MC SE	permutations	Holm p (substituted)	re
H4a	pre-registered run (p05, 10,000 permutations)	slot_sit	2833	20	303	388.4	1.0e-04	0.0001	10000	0.0007	Tr
H4b	pre-registered run (p05, 10,000 permutations)	slot_sit	2833	20	472	483.3	0.0221	0.0015	10000	0.0442	Tr
H4a	100,000 permutations, seed 0	slot_sit	2833	20	303	388.3	1.0e-05	0.0000	100000	7.0e-05	Tr
H4b	100,000 permutations, seed 0	slot_sit	2833	20	472	483.3	0.0236	0.0005	100000	0.0473	Tr
H4a	100,000 permutations, seed 1	slot_sit	2833	20	303	388.3	1.0e-05	0.0000	100000	7.0e-05	Tr
H4b	100,000 permutations, seed 1	slot_sit	2833	20	472	483.2	0.0236	0.0005	100000	0.0473	Tr
H4a	100,000 permutations,	slot_sit	2833	20	303	388.3	1.0e-05	0.0000	100000	7.0e-05	Tr

test	run	slot	tokens	streams	D	null mean	p	MC SE	permutations	Holm p (substituted)	re
	seed 2										
H4b	100,000 permutations, seed 2	slot_sit	2833	20	472	483.3	0.0233	0.0005	100000	0.0465	Tr
H4a	without hypocoristic and epithet forms	slot_sit	2616	20	254	282.3	5.0e-05	0.0000	20000	0.0003	Tr
H4b	without hypocoristic and epithet forms	slot_sit	2616	20	401	408.7	0.0646	0.0017	20000	0.1292	Fa
H4a	surname and first name only	slot_sit	2413	20	188	206.6	5.0e-05	0.0000	20000	0.0003	Tr
H4b	surname and first name only	slot_sit	2413	20	315	319.5	0.1232	0.0023	20000	0.2465	Fa
H4a	first reference per player per utterance	slot_sit	2171	20	280	350.1	5.0e-05	0.0000	20000	0.0003	Tr
H4b	first reference per player per utterance	slot_sit	2171	20	419	431.1	0.0097	0.0007	20000	0.0194	Tr
H4a	span-mode slot (classify(span) on the reference)	slot_sit_span	2833	20	305	389.8	5.0e-05	0.0000	20000	0.0003	Tr
H4b	span-mode slot (classify(span) on the reference)	slot_sit_span	2833	20	469	483.4	0.0055	0.0005	20000	0.0110	Tr
H4a	without the 3 streams with hypocoristics	slot_sit	1992	17	239	266.2	5.0e-05	0.0000	20000	0.0003	Tr
H4b	without the 3 streams with hypocoristics	slot_sit	1992	17	367	375.8	0.0443	0.0015	20000	0.0886	Fa

H4a: verdict **rejected (robust)** (rule: rejected (robust) iff the pre-registered run and every seed/variant are Holm-rejected under substitution); variants not Holm-rejected: none; leave-one-stream-out (20 runs, 5,000 permutations each): p 0.0002-0.0002, 20 below 0.05.

H4b: verdict **not robust / inconclusive** (rule: rejected (robust) iff the pre-registered run and every seed/variant are Holm-rejected under substitution); variants not Holm-rejected: without hypocoristic and epithet forms, surname and first name only, without the 3 streams with hypocoristics; leave-one-stream-out (20 runs, 5,000 permutations each): p 0.0104-0.0902, 17 below 0.05.

The streams whose omission raises the H4b p-value most: 2021 USO W F Raducanu-Fernandez (p 0.0902); 2022 RG M QF Djokovic-Nadal (p 0.0594); 2020 AO M F Djokovic-Thiem (p 0.0520). Commentary-only references by category: epithet 81, first_name 354, full_name 203, hypocoristic 136, surname 2059; hypocoristics occur in 3 streams (2019 WIM M F Federer-Djokovic, 2022 AO M F Nadal-Medvedev, 2022 RG M QF Djokovic-Nadal).

Table 4.5. Power of H4b (simulation: with probability theta a reference takes a designated form of its stream x player x slot cell; 200 data sets per theta, 499 permutations each).

theta	power at alpha 0.05
0	0.045
0.05	0.055
0.1	0.120
0.15	0.230
0.2	0.315
0.3	0.745
0.4	0.940

Table 4.5b. POST HOC (first round): H4 with de-clustered references, and the contribution of each slot to H4b (one reference per player per utterance; slot rows: distinct forms observed vs the mean under the H4b permutation, all commentary-only references; per-slot attribution rests on context-mode slots).

test	token set	tokens	D	null mean	null 95%	p (fewer)
H4a	all commentary-only references (as in H4)	2833	303	388.2	[377, 399]	1.0e-04
H4b	all commentary-only references (as in H4)	2833	472	483.2	[472, 494]	0.0268
H4a	first reference per player per utterance	2171	280	350.3	[340, 360]	1.0e-04
H4b	first reference per player per utterance	2171	419	431.1	[421, 441]	0.0100

slot	references	distinct forms (sum over stream x player)	null mean	observed - null	null 95%
CROWD	55	32	30.3	1.7	[27, 34]
JUDGE	535	89	94.3	-5.3	[87, 102]
OFFICIAL	37	20	19.6	0.4	[17, 23]
OTHER	1252	122	119.8	2.2	[113, 127]
SCORE	194	66	64.5	1.5	[58, 71]
SHOT	618	81	96.0	-15.0	[88, 104]
STAT	142	62	58.7	3.3	[53, 65]

Table 4.6. Extension: syllables vs available time (weighted mean of within-stream Spearman rho; permutation of the time values among utterances within stream; MDE = 2.80 x the null SD, normal approximation). The last row is post hoc (addendum 1).

analysis	references	time	tokens	streams	rho	stream bootstrap 95%	null 95%	p (two-sided)	MDE rho	st
H5	commentary_only	A_after (hit clock)	2811	20	0.028	[-0.012, 0.068]	[-0.036, 0.037]	0.1402	0.053	2b-p
H5 sensitivity	all_references	A_after (hit clock)	2933	20	0.025	[-0.016, 0.065]	[-0.037, 0.036]	0.1741	0.052	expl
H5 sensitivity	commentary_only	A_before (hit clock)	2821	20	0.009	[-0.032, 0.049]	[-0.037, 0.036]	0.6376	0.053	expl
H5 sensitivity	commentary_only	t_to_next_first_hit_s (stored; 25-fps streams only)	2205	14	0.029	[-0.013, 0.064]	[-0.042, 0.043]	0.1749	0.060	expl
E2 extension	E2 class occurrences (TV)	A_after (hit clock)	1881	255	0.018	[-0.039, 0.078]	[-0.063, 0.062]	0.5767	0.091	expl (stra stre clas
H5 sensitivity (post hoc)	commentary_only	t_to_next_first_hit_s (stored; all 20 streams after the corpus frame-rate correction)	2811	20	0.025	[-0.015, 0.062]	[-0.037, 0.037]	0.1891	0.053	post (add 1)

Mixed model (syllables ~ log(A_after) + (1 | stream), REML; 2811 references, 20 streams; exploratory): coefficient of log A_after = 0.014 syllables per log-second [-0.032, 0.059], p = 0.5610.

4.1 Stream-level reuse of formula expressions per slot (E1, exploratory)

Genre inventory identified on the 20 TV streams pooled: 23,110 formulas and 7,164 open-slot systems; greedy longest-first segmentation gives 30,777 occurrences. Per stream x slot (span-mode slot of the occurrence): occurrences, distinct types per 100 tokens of the slot and modal-type share; null: stream labels permuted among the slot's occurrences (10,000 permutations). With the pooled inventory a string repeated inside one stream only is in the inventory and all its occurrences belong to that stream, so 'fewer distinct types than chance' is guaranteed by construction; Table 4.7b repeats the test with each stream segmented by the inventory of the other 19 (post hoc). Even then the deficit is what within-document burstiness produces in any collection (a document reuses its own topical strings), so these tables measure stream-level reuse, not economy in Parry's sense (one expression per idea and metrical slot).

Table 4.7. E1 economy and across-stream null by slot (pooled inventory; rejects by construction).

slot	occurrences	distinct types per 100 slot tokens (mean over streams)	D (sum over streams of distinct types)	null D	p (fewer)	modal-type share % (mean over streams)	null	p (higher)
SCORE	2171	26.6	1946	1985.3	1.0e-04	6.5	7.2	0.9920
OFFICIAL	507	23.2	439	454.5	1.0e-04	16.7	13.8	0.0024
SHOT	7754	28.3	7419	7554.2	1.0e-04	0.9	0.7	0.0057
JUDGE	7208	28.2	6882	7018.9	1.0e-04	0.9	0.8	1.0e-04
STAT	2488	28.3	2379	2437.7	1.0e-04	2.5	2.0	0.0015
CROWD	634	23.8	610	624.9	1.0e-04	8.2	6.8	0.0026
OTHER	10015	19.6	9559	9762.8	1.0e-04	0.7	0.6	0.0009

Table 4.7b. POST HOC: E1 with a leave-one-stream-out inventory.

slot	occurrences	D	null D	null 95%	p (fewer)	modal-type share %	null	p (higher)
SCORE	2112	1842	1883.6	[1869, 1898]	1.0e-04	6.7	7.5	0.9979
OFFICIAL	496	376	409.8	[401, 418]	1.0e-04	23.4	17.5	1.0e-04
SHOT	7069	6455	6623.9	[6592, 6656]	1.0e-04	1.4	1.0	1.0e-04
JUDGE	6401	5882	5994.6	[5964, 6024]	1.0e-04	1.3	1.1	0.0006
STAT	2238	2055	2131.2	[2115, 2147]	1.0e-04	3.5	2.6	0.0002
CROWD	537	502	519.4	[513, 526]	1.0e-04	10.9	9.1	0.0028
OTHER	8993	8297	8483.7	[8447, 8520]	1.0e-04	1.0	0.8	1.0e-04

Commonest types per slot (pooled inventory, all streams): SCORE: `<num> <num>` 7.1% (19 streams), `<num> <num> <num> <num>` 1.0% (10 streams), `advantage <name>` 0.9% (9 streams), `out <num> <num>` 0.5% (3 streams); OFFICIAL: `thank you` 10.8% (15 streams), `thank you thank you` 1.6% (7 streams), `new balls` 1.6% (6 streams), `thank you very much` 1.2% (5 streams); SHOT: `drop shot` 0.2% (11 streams), `cross court` 0.2% (7 streams), `down the line` 0.2% (8 streams), `second serve` 0.1% (6 streams); JUDGE: `i think` 0.2% (11 streams), `you know` 0.2% (10 streams), `a little bit` 0.2% (7 streams), `under pressure` 0.2% (9 streams); STAT: `last year` 0.6% (10 streams), `this year` 0.4% (9 streams), `over the years` 0.4% (5 streams), `at times` 0.4% (6 streams); CROWD: `the crowd` 0.9% (4 streams), `in the box` 0.8% (4 streams), `the wind predictor` 0.8% (3 streams), `right now` 0.6% (4 streams); OTHER: `as well` 0.2% (10 streams), `for <name>` 0.2% (14 streams), `<num> <num>` 0.2% (11 streams), `of course` 0.1% (8 streams).

Table 4.8. Economy per stream and slot (E1, pooled inventory): distinct formula types per 100 tokens of the slot / modal-type share %.

stream	SCORE	OFFICIAL	SHOT	JUDGE	STAT	CROWD	OTHER
2019 WIM M F Federer-Djokovic	23.4 / 11	20.3 / 8	29.6 / 1	27.7 / 1	25.7 / 1	25.0 / 4	17.5 / 0
2023 WIM M F Alcaraz-Djokovic	25.0 / 12	21.8 / 3	28.9 / 1	29.1 / 1	27.5 / 2	24.0 / 4	19.0 / 1
2019 AO W F Osaka-Kvitova	22.9 / 7	28.6 / 10	28.5 / 1	27.4 / 1	27.9 / 1	23.8 / 4	18.8 / 1
2020 AO M F Djokovic-Thiem	25.5 / 6	20.5 / 6	26.8 / 0	28.3 / 1	29.2 / 1	22.4 / 3	20.2 / 1
2020 RG W F Swiatek-Kenin	29.0 / 3	25.0 / 22	29.7 / 2	29.8 / 2	25.0 / 12	23.1 / 33	21.9 / 1
2021 AO W F Osaka-Brady	30.2 / 3	31.0 / 11	28.8 / 1	27.7 / 1	28.4 / 5	23.0 / 7	20.6 / 1
2021 RG M F Tsitsipas-Djokovic	25.2 / 6	28.8 / 5	29.0 / 1	28.8 / 0	27.5 / 1	23.2 / 4	20.0 / 0
2021 USO W F Raducanu-Fernandez	26.4 / 5	20.3 / 29	29.9 / 1	26.1 / 1	28.4 / 2	24.1 / 13	19.8 / 1
2022 AO M F Nadal-Medvedev	24.8 / 7	22.1 / 33	28.6 / 0	27.3 / 1	26.8 / 1	22.9 / 5	19.6 / 1
2022 RG M QF Djokovic-Nadal	28.4 / 3	24.8 / 10	28.0 / 1	29.1 / 1	29.9 / 1	22.7 / 6	19.1 / 0
2022 USO M F Ruud-Alcaraz	27.4 / 4	19.0 / 12	26.8 / 1	26.9 / 1	29.8 / 2	23.6 / 7	19.9 / 1
2023 RG W F Swiatek-Muchova	26.7 / 6	22.2 / 21	28.2 / 1	28.0 / 1	27.6 / 3	30.0 / 8	20.2 / 1
2023 USO M F Djokovic-Medvedev	29.0 / 6	22.5 / 20	27.4 / 1	29.5 / 1	28.2 / 2	27.3 / 5	20.0 / 1
2024 AO M F Sinner-Medvedev	25.5 / 10	22.0 / 21	27.4 / 1	27.3 / 0	31.1 / 1	22.5 / 4	18.9 / 0
2024 RG W F Swiatek-Paolini	24.7 / 9	20.8 / 33	29.6 / 1	31.9 / 2	28.1 / 3	26.7 / 12	20.5 / 1
2024 WIM W F Paolini-Krejcikova	27.2 / 8	21.4 / 8	30.3 / 1	28.4 / 1	29.2 / 2	20.3 / 18	19.2 / 1
2024 USO M F Fritz-Sinner	28.2 / 6	25.0 / 17	26.4 / 1	27.6 / 1	28.4 / 2	23.0 / 4	19.2 / 1
2025 AO M SF Sinner-Shelton	26.8 / 5	31.4 / 9	28.8 / 1	27.6 / 1	27.4 / 3	26.0 / 3	19.8 / 1
2025 WIM W F Anisimova-Swiatek	30.3 / 5	16.0 / 25	26.9 / 1	28.0 / 1	29.2 / 2	17.1 / 17	17.7 / 1
2025 USO M F Sinner-Alcaraz	24.9 / 8	20.0 / 31	27.5 / 1	26.6 / 1	30.4 / 2	25.3 / 5	19.3 / 1

Slot share of all tokens (token-level classifier), mean over the 20 TV streams: SCORE 6.2%, OFFICIAL 1.6%, SHOT 22.0%, JUDGE 20.6%, STAT 6.9%, CROWD 2.0%, OTHER 40.7%.

4.2 Functional-equivalence classes (E2, exploratory)

Each class is a set of expressions declared equivalent in the plan (section 9) before counting. Forms pooled over the 20 TV streams, the Cornell and press modal forms, and the across-stream null (stream labels permuted among the class's TV occurrences; one-sided p for fewer distinct forms per stream / higher modal share than chance, i.e. stream-specific choice).

Table 4.9. Functional-equivalence classes.

class	slot	TV occurrences	TV streams using	TV forms (pooled)	Cornell modal form	press modal form	D / null D	p (fewer)	p (modal higher)
tied_at_forty	SCORE	40	14	deuce 98%, 40 40 2%	deuce (100%, n 423)	deuce (99%, n 216)	15 / 14.9	1.0000	0.9495
match_point	SCORE	35	14	championship point 51%,	match point	match point	17 / 20.3	0.0207	0.0139

class	slot	TV occurrences	TV streams using	TV forms (pooled)	Cornell modal form	press modal form	D / null D	p (fewer)	p (modal higher)
				match point 49%	(92%, n 40)	(100%, n 741)			
break_chance	SCORE	94	20	break point 90%, break chance 3%, break back point 2%, chance to break 2%	break point (93%, n 751)	break point (78%, n 594)	28 / 28.2	0.6045	0.9146
challenge	OFFICIAL	8	2	is challenging 100%	challenge from (85%, n 13)	going to challenge (54%, n 13)	/ -	-	-
net_error	SHOT	17	9	into the net 59%, in the net 24%, hits the net 18%	into the net (74%, n 538)	into the net (57%, n 170)	13 / 13.4	0.5232	0.4194
along_the_line	SHOT	93	17	down the line 77%, up the line 23%	down the line (92%, n 251)	down the line (79%, n 291)	23 / 27.6	0.0018	0.0156
serve_centre	SHOT	32	15	down the t 38%, up the middle 28%, down the middle 25%, down the centre 9%	down the middle (73%, n 102)	down the t (33%, n 52)	24 / 24.8	0.4031	0.4075
serve_body	SHOT	22	14	body serve 45%, into the body 45%, to the body 9%	to the body (44%, n 9)	to the body (61%, n 28)	16 / 18.0	0.0835	0.0465
serve_wide	SHOT	46	14	out wide 80%, wide serve 17%, serve wide 2%	out wide (89%, n 104)	out wide (79%, n 68)	18 / 20.3	0.0381	0.0030
small_degree	JUDGE	506	20	a little bit 37%, a little 35%, a bit 22%, slightly 7%	a bit (41%, n 171)	a little bit (63%, n 20103)	72 / 73.6	0.2474	0.0022
praise_stroke	JUDGE	85	19	good 35%, great 32%, what a 16%, beautiful 2%	superb (21%, n 311)	good (72%, n 1209)	50 / 51.9	0.2539	0.5498
well_played	JUDGE	31	11	well played 52%, well done 42%, beautifully played 6%	well played (67%, n 6)	well done (63%, n 121)	16 / 18.0	0.0824	0.0816
obligation	JUDGE	374	20	have to 43%, got to 28%, need to 23%, must 7%	have to (37%, n 272)	have to (62%, n 20181)	73 / 70.2	0.9676	0.5659
opinion_hedge	JUDGE	344	20	i think 83%, i feel 7%, i guess 5%, i suspect 2%	i think (58%, n 26)	i think (78%, n 51400)	53 / 56.8	0.0871	0.5718
speed_unit	STAT	33	13	miles an hour 67%, kilometres per hour 21%, miles per hour 12%	-	miles an hour (47%, n 86)	15 / 20.3	1.0e-04	1.0e-04
major_titles	STAT	29	15	grand slams 41%, majors	slams (96%, n	grand slams	22 / 23.5	0.2372	0.2731

class	slot	TV occurrences	TV streams using	TV forms (pooled)	Cornell modal form	press modal form	D / null D	p (fewer)	p (modal higher)
				31%, slams 21%, grand slam titles 3%	23)	(65%, n 1744)			
audience	CROWD	101	19	crowd 55%, fans 38%, audience 3%, supporters 3%	crowd (91%, n 99)	crowd (63%, n 2828)	39 / 39.7	0.4349	0.2660

Table 4.9b. POST HOC: E2 summed per slot (plan section 9 promised it; D summed over the slot's tested classes; null = independent within-class permutations, 10,000).

slot	classes	occurrences	D	null D	null 95%	p (fewer)
SCORE	tied_at_forty, match_point, break_chance	169	60	63.4	[60, 66]	0.0404
SHOT	net_error, along_the_line, serve_centre, serve_body, serve_wide	210	94	104.2	[99, 110]	0.0002
JUDGE	small_degree, praise_stroke, well_played, obligation, opinion_hedge	1340	264	270.6	[262, 279]	0.0722
STAT	speed_unit, major_titles	62	37	43.7	[40, 47]	0.0007
CROWD	audience	101	39	39.6	[37, 42]	0.4431

5. The seven 2b-primary tests (Holm-Bonferroni, alpha = 0.05)

Fixed in plan section 7 before any test ran; rerun unchanged. Null hypotheses: H1, matched-size coverage of a TV stream by another TV stream's inventory equals that of their unigram-shuffled texts; H2, that excess is no larger for TV sources than for the Cornell (H2a) / press (H2b) source; H4a, given the slot, the reference category is independent of the stream; H4b, within stream x player, the form is independent of the situational slot; H5, within stream, a reference's syllable count is independent of the time after its clip. H1/H2 p-values are percentile-bootstrap approximations over broadcasts treated as exchangeable draws; at their floor they only say that no bootstrap draw crossed zero. The last column gives the reading after the post hoc checks of addendum 1.

id	claim	estimate	interval	p	kind of p	Holm p	rejected	detail	reading after post hoc checks
H1a	TV teams share multiword stock beyond lexicon (n >= 2)	0.0756	[0.0699, 0.0823] (vertex bootstrap 95% CI)	0.0002	two-sided percentile-bootstrap, floor 0.00020	0.0012	True	20 of 20 TV targets positive	check only: 72% / 76% of the excess reached by press / Cornell
H1b	TV teams share multiword stock beyond lexicon (n >= 3)	0.0102	[0.0087, 0.0118] (vertex bootstrap 95% CI)	0.0002	two-sided percentile-bootstrap, floor 0.00020	0.0012	True	20 of 20 TV targets positive	check only; n >= 3 inventories at S = 1,500: 17-49 types
H2a	TV sources share more with TV targets than Cornell live text does	0.0184	[0.0142, 0.0233] (vertex bootstrap 95% CI)	0.0002	two-sided percentile-bootstrap, floor 0.00020	0.0012	True	20 of 20 TV targets positive	observed difference 0.53 [0.12, 1.01] pp (15/20); at 100,000 tokens 15.14 [14.29, 15.98] (n >= 2), 11.43

id	claim	estimate	interval	p	kind of p	Holm p	rejected	detail	reading after post hoc checks
									[10.70, 12.16] (n >= 3)
H2b	TV sources share more with TV targets than press answers do	0.0214	[0.0166, 0.0271] (vertex bootstrap 95% CI)	0.0002	two-sided percentile-bootstrap, floor 0.00020	0.0012	True	20 of 20 TV targets positive	observed difference 3.10 [2.46, 3.84] pp (20/20); at 100,000 tokens 13.09 [11.85, 14.22] (n >= 2), 11.33 [10.36, 12.23] (n >= 3)
H4a	naming habits (reference categories) are team-specific	303	null mean 388.4, 95% [377, 399]	1.0e-04	one-sided permutation (fewer than chance), 10000 permutations	0.0007	True	2833 commentary-only references	rejected (robust)
H4b	form choice is bound to the situational slot within team x player	472	null mean 483.3, 95% [473, 494]	0.0221	one-sided permutation (fewer than chance), 10000 permutations	0.0442	True	2833 commentary-only references; MDE theta at 80% power 0.328	not robust / inconclusive (p 0.0055-0.1232 over seeds and variants; speaker-role confound untestable)
H5	longer referring expressions where more time is available	0.0278	stream-bootstrap 95% CI [-0.012, 0.068]; null 95% [-0.036, 0.037]	0.1402	two-sided permutation, 10000 permutations	0.1402	False	2811 references in 20 streams; MDE rho about 0.053	stored time on all 20 streams: rho 0.025, p 0.1891

The p-values of H1a, H1b, H2a, H2b are at the bootstrap floor (2/(B+1)).

Exploratory p-values reported in sections 2 and 4 (counted per p-value, not per row): 125 from the planned exploratory analyses (cluster_tests.csv 12; qap_regression.csv (incl. intercepts) 12; thrift_tests_2b.csv (non-primary rows) 44; extension_2b.csv (non-primary rows) 4; mixed model (primary_H4H5.json) 1; e1_tests.csv (2 per slot) 14; e2_tests.csv (2 per class) 32; h4_declustered.csv (first-round post hoc) 4; Spearman, stream-specific share vs length (core_meta.json) 2) and 76 from the post hoc checks of addendum 1 (h4_robust.csv (seeds and variants) 16; h4_robust_loso.csv 40; e1_loso_tests.csv (2 per slot) 14; e2_slot_sums.csv 5; h5_sensitivity_post_hoc.csv 1). All are unadjusted; at alpha 0.05 about 10 would fall below 0.05 by chance if every null held. The bootstrap p-values stored in `h2_observed.csv` are not used; differences there are reported with intervals.

6. Limitations

1. **Repetition, not Parry's formula.** Every inventory is a set of exact n-grams repeated within a text. Against two LLM coders applying Parry's definition, the automatic measures reach kappa 0.11-0.22 (coders with each other 0.77); see section 3. Agreement with human specialists is untested.
2. **ASR noise.** The TV texts are raw WhisperX output with an unknown word error rate (lower bound 22 visible errors per 1,000 words, Phase 2); errors break exact repeats in both the source and the target of a TV-TV comparison but only in the

target of a Cornell/press-TV comparison, which biases the TV-minus-baseline differences toward zero. The press answers are edited stenography and Cornell is edited prose.

3. **Clip-level text and time.** A clip's text runs into the following dead time; slots are assigned to spans or windows by a keyword classifier, and A_after is a clip property. Extension and thrift tests are about clip-level co-variation, not the moment a word is chosen.
4. **Streams are not teams.** A stream is one match with one unnamed broadcaster; commentator hints exist for four streams and are [unverified]; the only two-member hint cluster coincides with same slam, same player and same round. Same-slam similarity mixes broadcaster, venue and surface vocabulary.
5. **No speaker labels.** Two or more commentators speak in a stream; a play-by-play voice and an analyst voice that differ in naming and in the situations they speak in produce the same patterns as slot-conditioned choice (H4b) or as stream-level habits (H4a, E2).
6. **Slot labels are noisy.** Context mode, which gives a reference its slot in H4, agrees with the coders at kappa 0.35 [0.22, 0.49] / 0.39 [0.25, 0.52] on the 65 / 69 commentary references inside coded spans, where the coder slot is that of the covering span (a proxy: no coder labelled reference windows); span mode (formula spans) at 0.49 / 0.48, with OTHER as a residual class. Random slot errors attenuate H4b and blur the per-slot tables; systematic ones bias compositions: the TV-only composition is given as classified (Table 2.E) and corrected (Table 2.E2), and the correction depends on which conditional probabilities are assumed to carry over from coder spans to TV-only strings. Tables 2.11, 2.E and 4.7 use span mode on the strings themselves, so their SCORE and OFFICIAL rows are partly defined by the strings' own words.
7. **Sizes and samples.** At 1,500 tokens an n >= 3 inventory has 17-49 types, too few for n >= 3 comparisons; the 100,000-token design fixes that, but its TV source is drawn from 19 streams that overlap between targets, so its bootstrap over targets understates between-broadcast uncertainty, and the 5 TV samples of a target overlap by most of their tokens (Table 2.A). 'TV-only' means 'repeated in a 100,000-token TV sample and not in 100,000-token Cornell and press samples'; such strings occur in the baselines at rates a median 8.4 times lower (Table 2.F2), and the 'strict' class tests a 100,000-token TV sample against whole baselines 1.8 and 54 times larger. Part of the TV advantage is same-slam and same-player sharing (Table 2.H).
8. **Baselines are different corpora.** Cornell (one outlet, Sports Mole live text (per Cornell dataset README; written, not speech) [source site not independently verified]; written; other matches; no dates in the records) and the press answers (2007-2015; player answers, first person) differ from the TV streams (2019-2025) in matches, period, transcription, segmentation and person; TV-minus-baseline differences say that TV commentary repeats strings these sources do not, not why. Name normalisation is uneven: hypocoristics are not normalised, and `<name>` is rare in press (section 1).
9. **Hand inputs.** The slot lexicons, the E2 equivalence classes and the demonym lists are my own specification; three nationalities (Alcaraz, Raducanu, Shelton) are entered by hand [unverified]; descriptive epithets outside the two finals are left unresolved rather than hand-assigned.
10. **Coders.** Both codings come from LLM agents (git trailers in section 3); the coding instructions in the repository are `manual.md`, and the agents' task prompts are not recorded there; their blindness to the automatic inventory rested on those instructions.
11. **Exploratory status.** All of 2b was designed after Phase 2's results were known; the 2b-primary family limits multiplicity within 2b but is not a pre-registration in the Phase 2 sense; everything in addenda 1 and 2 is post hoc and was designed after the critic's reruns had been read.
12. **Not done.** Coders' slot labels for masked reference windows (the direct validation of the context-mode slot asked for in both reviews); a yardstick for E1 on Cornell or press documents; TV samples drawn small enough to be independent replicates (the overlap is stated instead, Table 2.A); a selection-free estimate of the TV / baseline rate ratio (strings chosen on one half of the TV streams and rated on the other).

7. Post hoc changes (made after a first look at outputs)

First round (before the review):

1. Nationality epithets: a match whose demonym modifies a following non-person noun ('the greek fans', 'the czech republic', 'the italian riviera') is not counted as a player reference (spaCy dependency rule; 17 matches excluded, each listed with its excerpt in `results/refexpr_pool_epithets.csv`; a few exclusions are parser errors, e.g. 'the pole powers on'). The outputs of the run before this change were not kept.
2. Hypocoristic + surname ('rafa nadal', 'nole djokovic') counted as one full-name reference (6 cases), not two (first-run outputs not kept).
3. Added beyond the plan's text: the full-size inventory table includes Cornell but not the press corpus (too large to index whole); the E2 extension uses 2000 permutations (it has about 300 strata); `title_surname` is a sixth reference category beside the plan's five (it occurs only inside umpire patterns, so it is absent from the commentary-only tests). 3b. Calibration: the plan's 2019-versus-other split is reported for overall precision / recall / F1 (Tables 3.2b), not within each slot or confidence stratum.
4. H4b power simulation corrected: the first run planted slot-specific forms on the observed data, which already carries the observed effect, so its 'power' at theta = 0 was the rejection rate of the observed data (0.995); the corrected simulation starts every data set from the observed forms with slot labels permuted within stream x player, so H0 holds at theta = 0 (Table 4.5).
5. Table 4.5b (H4 with one reference per player per utterance, and the per-slot contribution to D_b) was added after the H4b result was seen.

Revision 1 (plan addendum 1, after `review/critic_parry_v1.md`; all post hoc):

6. Large identification sets (100,000 tokens; `p08_largeI.py`) became the primary cross-corpus comparison, with the commentary-only variant, the TV-only n >= 3 stock, its strict and cross-broadcast readings, its slot composition and its string lists (section 2.0). The first report's statement about n >= 3 units rested on the 1,500-token inventories that this design replaces.
7. H2 is reported on observed coverage beside the pre-registered excess (Table 2.3, `p09_h2_observed.py`); Table 2.3b explains the size of the Cornell null; H1 is reported as a check.
8. Calibration at chance level, kappa of every automatic measure, span-level recall against circular shifts, and the slot classifier in context mode on coder spans and on references (`calibrate_chance.py`; `calibrate.py` now also saves its automatic labels).
9. H4a/H4b robustness: three seeds at 100,000 permutations, five variants, leave-one-stream-out, Holm under substitution, and a verdict rule (`p10_h4_robust.py`); `p04_refexpr_pool.py` writes two more columns (`slot_sit_span`, `t_next_stored_all`); H5 with the stored time field on all 20 streams.
10. E1 with a leave-one-stream-out inventory and E2 summed per slot (`p11_e1_loso.py`); p-values counted per p-value; 'idiolect' renamed 'stream-specific share'; LOW-confidence rows dropped from Tables 3.1 and 3.3; the rerun check against commit 476a088 (`p12_rerun_check.py`).
11. The plan declares 17 E2 classes; one (`challenge`) has a single attested TV form and no test, so 16 are tested.

Revision 2 (plan addendum 2, after `review/critic_parry_v2.md`; all post hoc):

12. Rates of the TV-only strings at matched sample size (`p13_rate_ratio.py`; Key results 'Rarer elsewhere, not absent', Table 2.F2). The sentence 'most of the TV-only stock consists of strings the other registers also use, less often (a difference of frequency)', which rested on the asymmetric 'strict' test and gave no magnitude, is replaced by the matched-size rate ratio; the strict class is labelled asymmetric. The planned bootstrap over replicates was replaced by the two halves of the replicates and a bootstrap over strings, because resampling replicates loses the attestation of rare strings (its intervals excluded their own estimates).
13. Slot composition of the TV-only stock corrected for the classifier's confusions, with two estimators (`p14_slot_adjust.py`; Table 2.E2, fig10); the composition rule now uses

the corrected share; Table 2.E calls OTHER a residual class.

14. `p08_largeI.py` writes the full TV-only and shared string lists, an umpire-mask check and a composition variant with the umpire's game announcements as OFFICIAL (Table 2.E); its earlier outputs are byte-identical. The check found that `official_mask_v2` already contains the umpire's game announcements (it applies the Phase 2 mask, which has `<name> leads (by) N games/sets to M`, first): 32 of 42 such frames in the TV texts lie wholly inside it, the others lack a recognised player name (ASR) or are commentators' sentences, so the commentary-only rows already exclude the announcements and the mask is unchanged.
15. Same-slam and same-player sensitivity with a random-removal control (`p15_sameslam.py`; Table 2.H).
16. The context-mode kappa on references is given with a bootstrap interval (`p14_slot_adjust.py`); the coder-span context-mode kappa left the Key results and limitation 6 (Table 3.4 explains why it is not an error rate).
17. E1 is called stream-level reuse, with the burstiness explanation (Key results, section 4.1); the title says 'repetition across broadcasts'; Table 2.A states the overlap of the TV samples; Table 2.F notes the full-name habit; the revision paragraphs say which rules were fixed after first outputs.

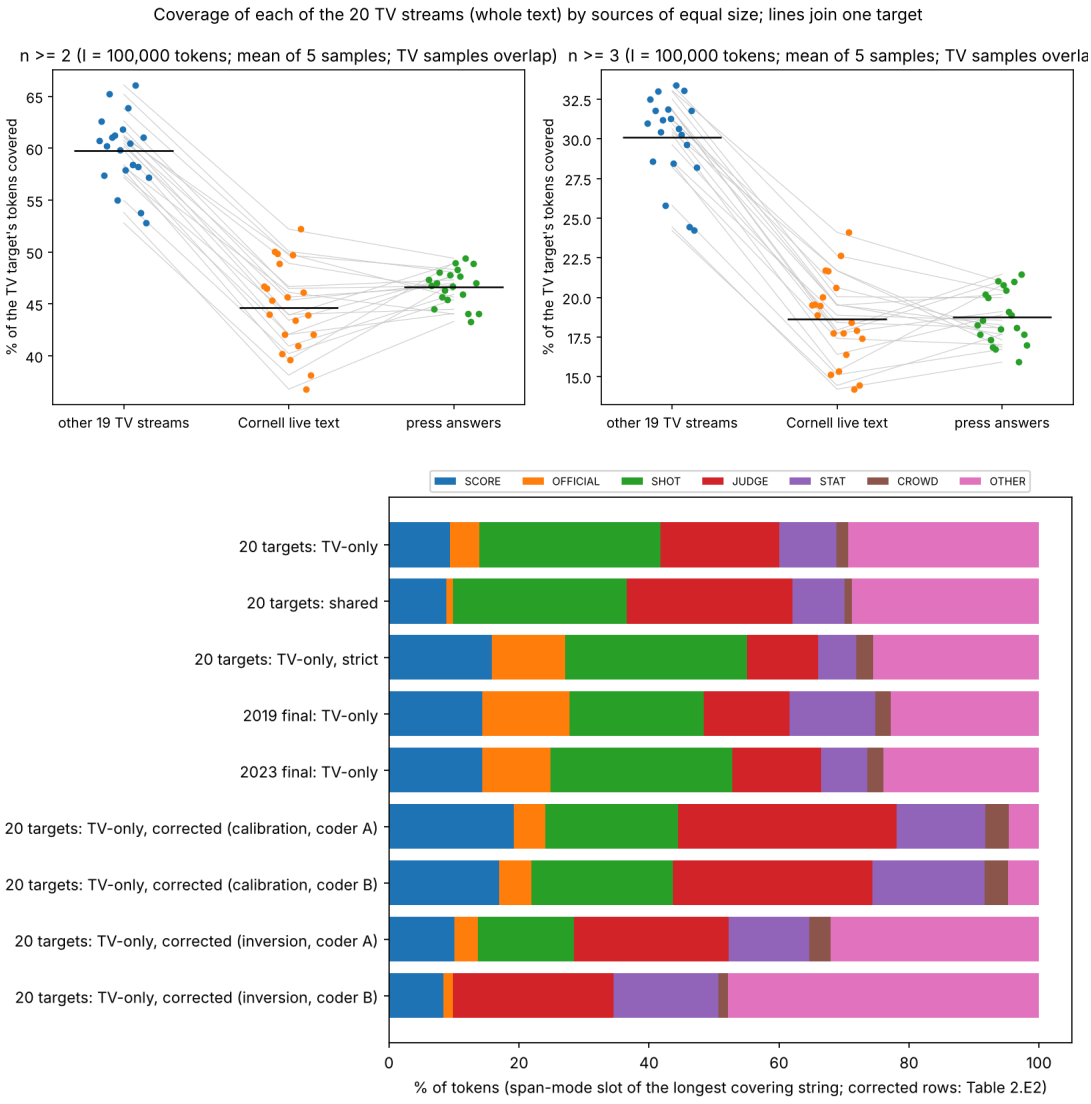
Verdict wording rules (thresholds, when each was fixed, observed values, and the larger of value/threshold and threshold/value):

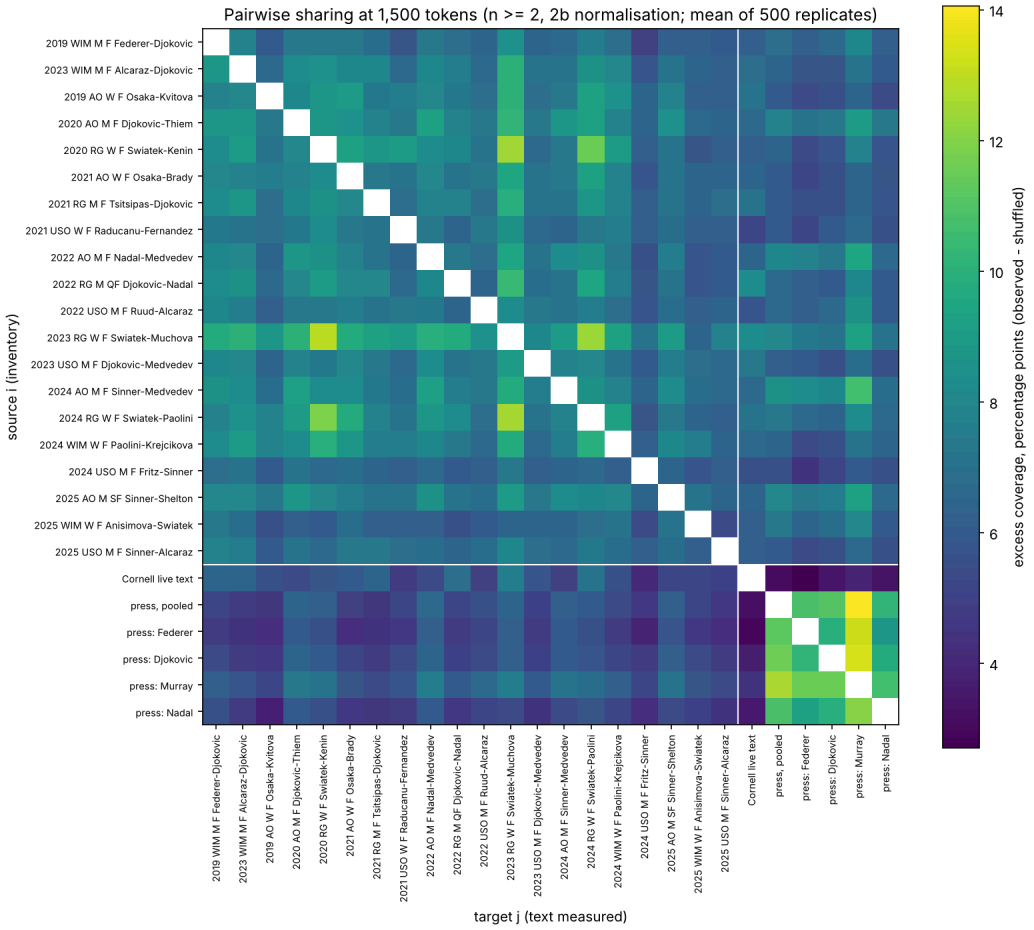
rule	threshold	fixed in	observed	ratio
near-empty n >= 3 inventories: TV -> TV n >= 3 coverage at S = 1,500 below	0.020	addendum 1 design (before its outputs)	0.010	1.96
weak proxy: largest kappa of an automatic measure below	0.400	addendum 1 design (before its outputs)	0.217	1.85
slot-noise caveat: largest classifier kappa below	0.600	addendum 1 generator / implementation notes (after the first outputs)	0.561	1.07
base rate:	difference of precision - chance	below half the raw difference	0.082	addendum 1 generator / implementation notes (after the first outputs)
hint cluster: 'no sign' if p at least	0.050	addendum 1 generator / implementation notes (after the first outputs)	0.232	4.63
composition: upper bound of the corrected SCORE + STAT + OFFICIAL share below	0.500	addendum 2 (before its outputs)	0.431	1.16
rate wording: median TV / max(baseline) ratio at least 10 (else floor)	10.000	addendum 2 (before its outputs)	8.360	1.20
rate wording: share of TV-only occurrences unattested in both baselines below	0.500	addendum 2 (before its outputs)	0.164	3.05
share_word: cross-broadcast part of the TV-only tokens (nearest band limit)	0.900	addendum 1 generator / implementation notes (after the first outputs)	0.934	1.04
share_word: H1 excess reached by the baselines (nearest band limit)	0.900	addendum 1 generator / implementation notes (after the first outputs)	0.717	1.25

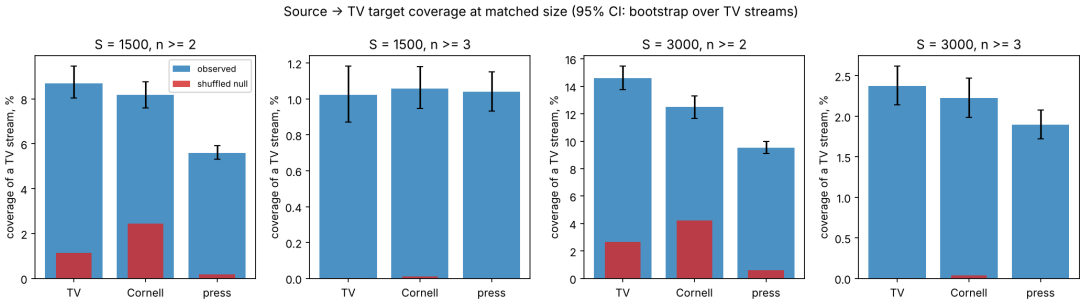
Observed values within a factor of 1.5 of their threshold: slot-noise caveat (0.561 vs 0.600); composition (0.431 vs 0.500); rate wording (8.360 vs 10.000); share_word (0.934 vs 0.900); share_word (0.717 vs 0.900). A different threshold could change those words; the numbers beside them are printed in every case.

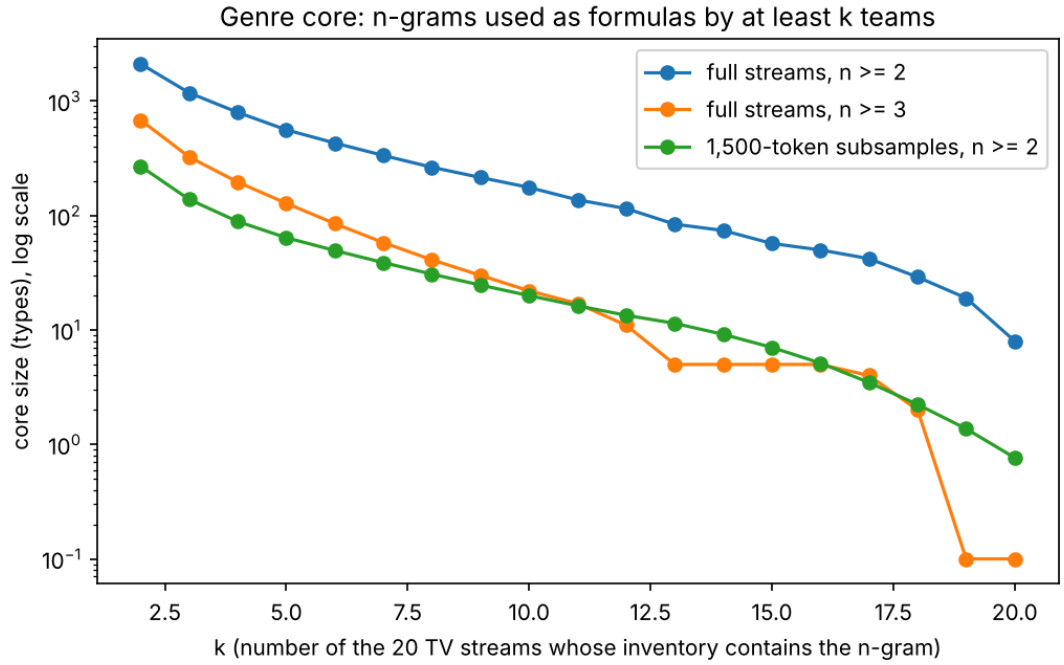
8. Files

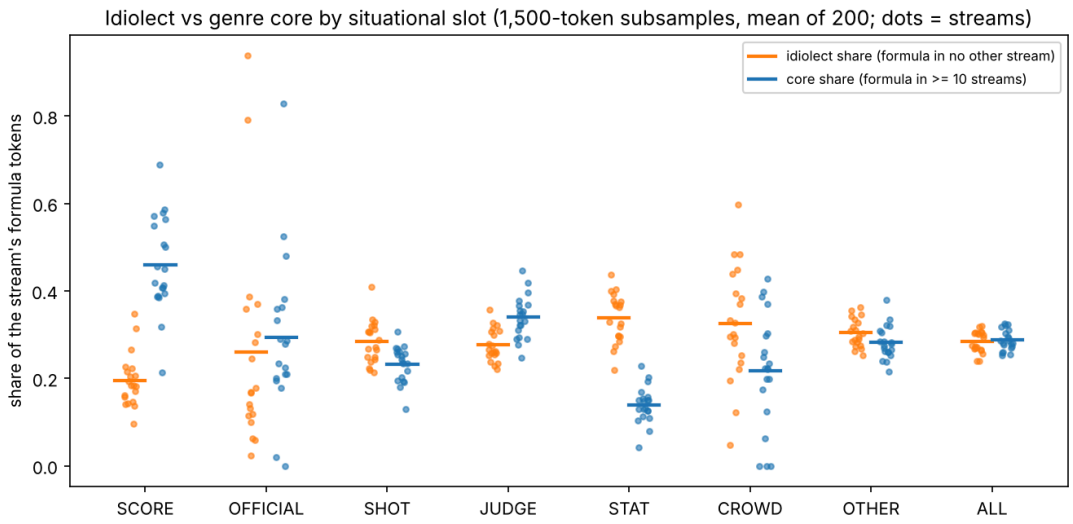
Code: `lib2b.py`, `p00_players.py`, `p01_sharing.py`, `p02_core.py`, `p03_clusters_targets.py`, `p04_refexpr_pool.py`, `p05_thrift.py`, `p05b_h4_declustered.py`, `calibrate.py`, `p06_primary.py`; post hoc (addendum 1): `p08_largeI.py`, `p09_h2_observed.py`, `calibrate_chance.py`, `p10_h4_robust.py`, `p11_e1_loso.py`, `p12_rerun_check.py`; post hoc (addendum 2): `p13_rate_ratio.py`, `p14_slot_adjust.py`, `p15_sameslam.py`; `p07_figures.py`, `make_report.py`, `run_all.sh`, `tests/`. Hand input: `hand/players.tsv`. Results: `results/*.csv|json`. Figures: `figures/fig1_sharing_heatmap.png` … `fig8_calibration.png`; post hoc `fig9_largeI_coverage.png`, `fig10_tvonly_slots.png`.

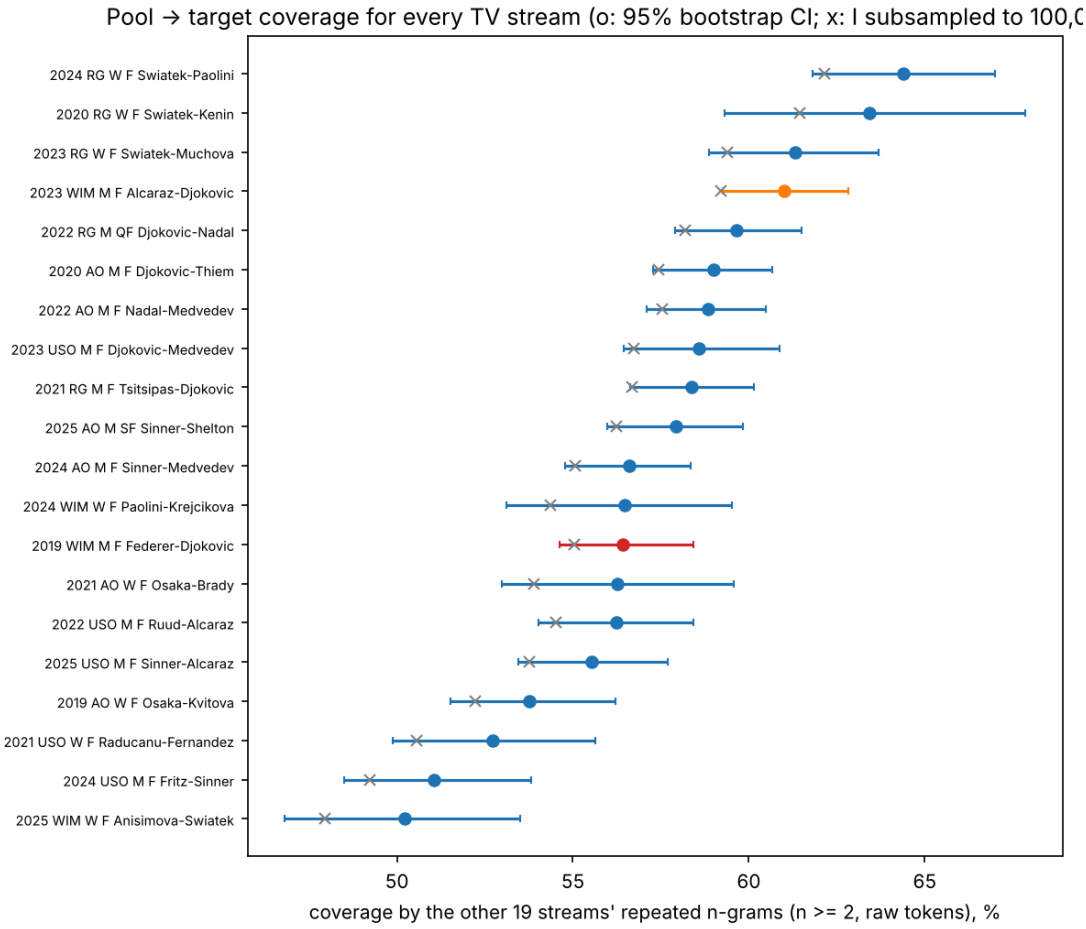


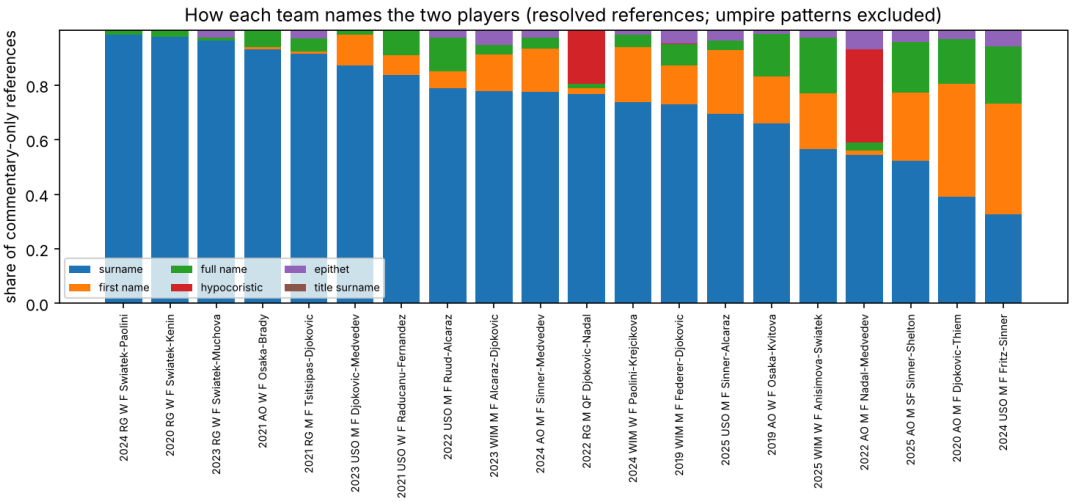

Pairwise sharing at 1,500 tokens (n >= 2, 2b normalisation; mean of 500 replicates)
source i (inventory)
target j (text measured)
excess coverage, percentage points (observed - shuffled)
2019 WIM M F Federer-Djokovic
2023 WIM M F Alcaraz-Djokovic
2019 AO W F Osaka-Kvitova
2020 AO M F Djokovic-Thiem
2020 RG W F Swiatek-Kenin
2021 AO W F Osaka-Brady
2021 RG M F Tsitsipas-Djokovic
2021 USO W F Raducanu-Fernandez
2022 AO M F Nadal-Medvedev
2022 RG M QF Djokovic-Nadal
2022 USO M F Ruud-Alcaraz
2023 RG W F Swiatek-Muchova
2023 USO M F Djokovic-Medvedev
2024 AO M F Sinner-Medvedev
2024 RG W F Swiatek-Paolini
2024 WIM W F Paolini-Krejcikova
2024 USO M F Fritz-Sinner
2025 AO M SF Sinner-Shelton
2025 WIM W F Anisimova-Swiatek
2025 USO M F Sinner-Alcaraz
Cornell live text
press, pooled
press: Federer
press: Djokovic
press: Murray
press: Nadal

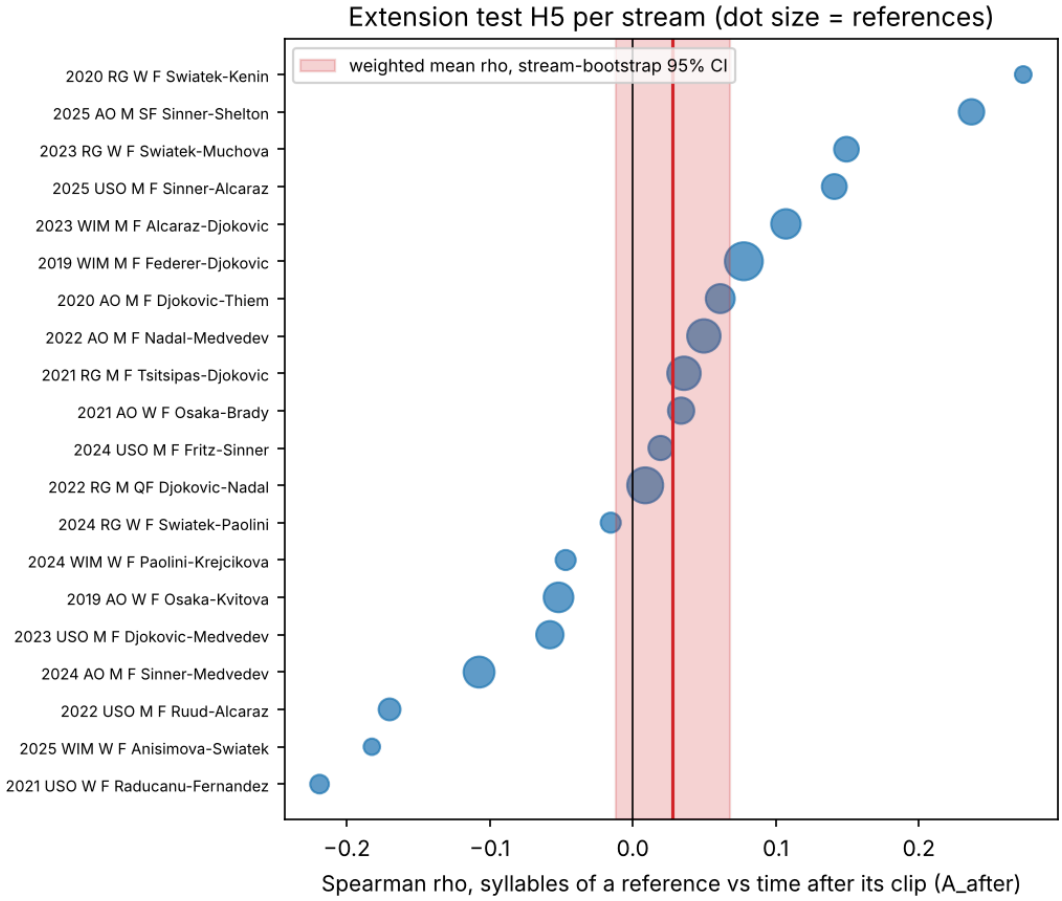

Source → TV target coverage at matched size (95% CI: bootstrap over TV streams)
S = 1500, n >= 2
S = 1500, n >= 3
S = 3000, n >= 2
S = 3000, n >= 3
observed
shuffled null
coverage of a TV stream, %
TV
Cornell
press

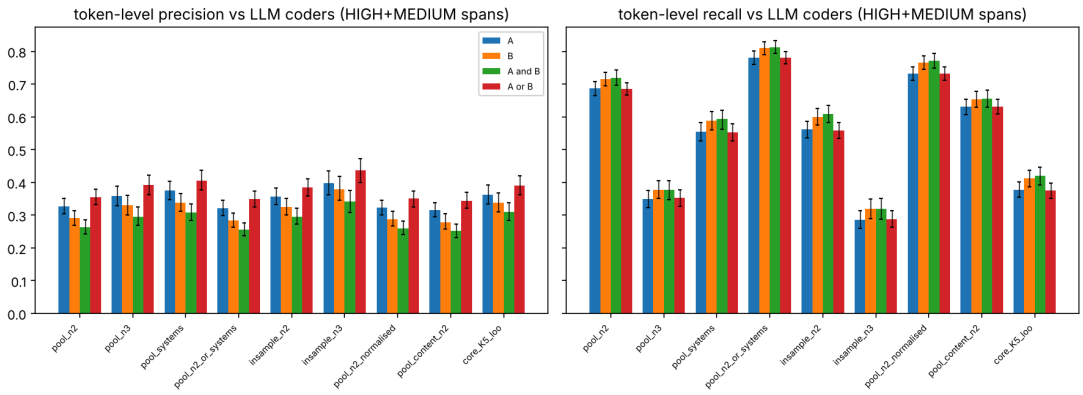

Genre core: n-grams used as formulas by at least k teams
full streams, n >= 2
full streams, n >= 3
1,500-token subsamples, n >= 2
core size (types), log scale
10^3
10^2
10^1
10^0
10^{-1}
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
k (number of the 20 TV streams whose inventory contains the n-gram)


Idiolect vs genre core by situational slot (1,500-token subsamples, mean of 200; dots = streams)
idiolect share (formula in no other stream)
core share (formula in >= 10 streams)
share of the stream's formula tokens
0.0
0.2
0.4
0.6
0.8
SCORE
OFFICIAL
SHOT
JUDGE
STAT
CROWD
OTHER
ALL


Pool → target coverage for every TV stream (o: 95% bootstrap CI; x: I subsampled to 100,(
2024 RG W F Swiatek-Paolini
2020 RG W F Swiatek-Kenin
2023 RG W F Swiatek-Muchova
2023 WIM M F Alcaraz-Djokovic
2022 RG M QF Djokovic-Nadal
2020 AO M F Djokovic-Thiem
2022 AO M F Nadal-Medvedev
2023 USO M F Djokovic-Medvedev
2021 RG M F Tsitsipas-Djokovic
2025 AO M SF Sinner-Shelton
2024 AO M F Sinner-Medvedev
2024 WIM W F Paolini-Krejcikova
2019 WIM M F Federer-Djokovic
2021 AO W F Osaka-Brady
2022 USO M F Ruud-Alcaraz
2025 USO M F Sinner-Alcaraz
2019 AO W F Osaka-Kvitova
2021 USO W F Raducanu-Fernandez
2024 USO M F Fritz-Sinner
2025 WIM W F Anisimova-Swiatek
50
55
60
65
coverage by the other 19 streams' repeated n-grams (n >= 2, raw tokens), %


How each team names the two players (resolved references; umpire patterns excluded)
share of commentary-only references
0.0
0.2
0.4
0.6
0.8
surname
first name
full name
hypocoristic
epithet
title surname
2024 RG W F Swiatek-Paolini
2020 RG W F Swiatek-Kenin
2023 RG W F Swiatek-Muchova
2021 AO W F Osaka-Brady
2021 RG M F Tsitsipas-Djokovic
2023 USO M F Djokovic-Medvedev
2021 USO W F Raducanu-Fernandez
2022 USO M F Ruud-Alcaraz
2023 WIM M F Alcaraz-Djokovic
2024 AO M F Sinner-Medvedev
2022 RG M QF Djokovic-Nadal
2024 WIM W F Paolini-Krejcikova
2019 WIM M F Federer-Djokovic
2025 USO M F Sinner-Alcaraz
2019 AO W F Osaka-Kvitova
2025 WIM W F Anisimova-Swiatek
2022 AO M F Nadal-Medvedev
2025 AO M SF Sinner-Shelton
2020 AO M F Djokovic-Thiem
2024 USO M F Fritz-Sinner


Extension test H5 per stream (dot size = references)
weighted mean rho, stream-bootstrap 95% CI
2020 RG W F Swiatek-Kenin
2025 AO M SF Sinner-Shelton
2023 RG W F Swiatek-Muchova
2025 USO M F Sinner-Alcaraz
2023 WIM M F Alcaraz-Djokovic
2019 WIM M F Federer-Djokovic
2020 AO M F Djokovic-Thiem
2022 AO M F Nadal-Medvedev
2021 RG M F Tsitsipas-Djokovic
2021 AO W F Osaka-Brady
2024 USO M F Fritz-Sinner
2022 RG M QF Djokovic-Nadal
2024 RG W F Swiatek-Paolini
2024 WIM W F Paolini-Krejcikova
2019 AO W F Osaka-Kvitova
2023 USO M F Djokovic-Medvedev
2024 AO M F Sinner-Medvedev
2022 USO M F Ruud-Alcaraz
2025 WIM W F Anisimova-Swiatek
2021 USO W F Raducanu-Fernandez
−0.2
−0.1
0.0
0.1
0.2
Spearman rho, syllables of a reference vs time after its clip (A_after)


token-level precision vs LLM coders (HIGH+MEDIUM spans)
A
B
A and B
A or B
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
pool_n2
pool_n3
pool_systems
pool_n2_or_systems
insample_n2
insample_n3
pool_n2_normalised
pool_content_n2
core_K5_loo
token-level recall vs LLM coders (HIGH+MEDIUM spans)
pool_n2
pool_n3
pool_systems
pool_n2_or_systems
insample_n2
insample_n3
pool_n2_normalised
pool_content_n2
core_K5_loo